

THE ENGINEERING EQUIPMENT AND MATERIALS USERS' ASSOCIATION

PROCESS PLANT CONTROL DESKS UTILISING HUMAN-COMPUTER INTERFACES

A GUIDE TO DESIGN, OPERATIONAL AND HUMAN INTERFACE ISSUES

PUBLICATION N° 201 : 2002

EEMUA

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**10-12 Lovat Lane
London
EC3R 8DN
Tel + 44(0) 20 7621 0011
Fax + 44(0) 20 7621 0022
sales@eemua.org
www.eemua.org**





THE ENGINEERING EQUIPMENT AND MATERIALS USERS' ASSOCIATION

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Foreword

In years past, the board operators of large industrial processes would perform monitor and control activities from large panelboards containing scores of single loop controllers and indicators with trend charts and alarm enunciators provided for a limited number of key variables. Because of the cost and limited size of the panelboards, project engineers spent considerable effort designing content and layout before implementation. Once built, operators learned the content, where to find things, and a feel for 'normal'. At a glance, an operator could get a sense of the overall process status from just about anywhere in the room. Navigation from one part of the unit to another was not difficult to learn—one simply walked from one location to another.

Modern DCS control systems have been justified based on improving control and optimisation, increasing throughput, controlling at constraints, and increasing the board operator's scope of control. Consciously or not, demands of operator capability and performance have significantly changed; indeed the job of operators has transitioned from one of considerable reliance on knowledge of the plant layout and memorisation of a number of tasks to one more focused on process knowledge and cognitive skills.

Humans are not particularly well suited for watching computer monitors hour after hour, waiting for something to happen...which then rarely does. Plant managers and engineers did not fully understand the human factors implications of the introduction of this new technology into the control room environment. Inadequately designed graphics for the operators in many cases leads to significantly diminished situational awareness and operator fatigue.

The persistent paradox in the process industry is that as automation technology increases in complexity and sophistication, operators are faced with increasingly complex decisions in managing the process. Moreover, the industry's ability to design effective operator interfaces to the process has not kept pace with the task demands imposed by increased sophistication of technology. A key factor is the ability to understand the operator's task and interaction requirements and to design the elements of the work environment that effectively support those requirements. Hence, we have entered an era in which there is general consensus throughout the industry that the design of the operator interface, console and control room deserves appropriate skills and resources.

Moreover the advent of new user interface technology has complicated the situation even further: "Use of sophisticated technology does not inherently increase system effectiveness; increases in effectiveness depend on how technology is used. Technology increases design flexibility, which in turn creates opportunities to dramatically enhance system performance. Flexibility however, also creates potential pitfalls and traps for the naive or unwary. A design may enhance system effectiveness by enhancing the effectiveness of operations personnel and mitigating the consequences of human error. Interfaces and human-system interaction design in sophisticated systems, however, have frequently been identified as major contributors to poor operator performance which sometimes results in dangerous or costly incidents and accidents." (Mitchell & Sundstrom, 1997¹)

The process industry is beginning to understand that the development of sub-optimal operator interfaces and work environments can have an impact on plant safety and profitability. People experienced in plant operations and control created this guideline

¹ Mitchell, C. and Sundstrom, G. (1997). *IEEE Transactions on Systems, Man, and Cybernetics: Special Issue on Human Interaction with Complex Systems*, 27(3), p. 265.

document. Grounded in their learning over the past decade, this document provides an excellent overview of the issues that plant management, project personnel and plant technical staff should be aware of as they make decisions regarding the design of the operator interface, console and work environment. The Abnormal Situation Management (ASM®) Consortium is pleased to recommend this document to those wishing to improve the effectiveness of console operations.

Michael J Clark
for the ASM® Executive Steering Committee

About this Guide

Scope

This document relates to Human Computer Interface (HCI) systems provided for people operating industrial processes such as chemical plant, power stations, oil refineries etc. It gives guidance on:

- the factors to take into account when designing an HCI for such processes;
- display hierarchies;
- screen display format design;
- control room design.

The objective is to help engineers to develop HCI systems which are more usable and which therefore result in safer and more cost-effective operation of industrial systems. In addition it is intended to help reduce the need for re-engineering of systems after installation.

Purpose

HCI systems now form the essential operator interface to large modern industrial systems. They are utilised in power stations, refineries, chemical plants (continuous and batch), utility distribution centres, and many other places. They provide the vital means by which the operator obtains information on the state of the plant, enters operational parameters, and by which any automatic control action can be overridden and manual control of the plant be taken.

The purpose of this Guide is to help those involved in the design, procurement, operation, maintenance and management of these systems. It is intended to help both in improving existing systems and in developing new facilities during plant construction. It is for use by engineers and managers in both user and contracting organisations. The objective is to make plants more operable and to save money.

This Guide has been written by users in industry. It is based on what some leading companies are doing. The guidance is designed to be practical and usable. The general guidance is directed towards large sophisticated computer-based process systems, but contains much general information on HCI. There is also detailed guidance on specific issues.

Acknowledgements

This Guide has been prepared by an EEMUA Sub-Committee made up of representatives of EEMUA member companies.

A list of EEMUA Member Companies who have been involved with preparation and review of this and other EEMUA publications is included at the end of the book.

EEMUA gratefully acknowledges the contributions of all who have been involved, in many capacities, during the preparation and production of this Guide.

Notes on use of this Guide

The main body of the text concentrates on philosophy and general principles. Key messages are highlighted in **bold**. Much supporting detail is in the appendices.

A glossary defining a few terms that readers may find unfamiliar or may cause confusion is given in Appendix 1. References are listed in Appendix 7. They are referred to in parentheses, e.g. (ref. 1).

1. An introduction to computer based HCI control rooms

Section overview

This section provides a basic introduction to HCI control rooms.

1.1 What is a HCI system?

Computer screen based systems now pervade every aspect of daily living, from the cash-point machine to the airline booking system and process control is no exception.

In the past most control rooms consisted of hard-wired equipment (such as dials, chart recorders, push buttons and knobs) laid out on large metal panels and desks. In some industries the operator would patrol the panels monitoring key plant variables, adjusting set-points and operating equipment. In other industries the majority of the work was performed at the desk, with the panels being used mainly for observation purposes. This type of hard-wired operation meant that all the indicators that an operator needed were immediately visible and that in periods of high workload, the control operations could be easily shared.

Evolution over the years in most industries had seen control rooms reach a point of relatively high sophistication and where ergonomic guidelines had been developed for all aspects of the panel & desk layout, from labelling to colour coding.

With the advent of computer operated control systems, a further desk development had taken place, namely the introduction of computer screens to allow for viewing of plant parameters, via numeric indication and graphics (trends and display formats). These screens were usually seen as an 'add-on' facility - they did not allow for operator control action and had to be integrated in to the existing desk design.

The final phase of the control desk transition has now taken place in the process industry. This is the provision of a full 'soft-desk', comprising completely of computer based screens, through which the operator both views the plant and operates it. In the majority of cases, there is no hard-wired indication facility at all (the exception being sometimes an Emergency Shutdown system). This then is the HCI. Where new plant is built such systems are now invariably installed. All the major manufacturers and independent DCS suppliers offer these as standard and many plants are refurbishing their control rooms and replacing the panels & hard-desk with a soft-desk.

The issue that has to be addressed by design engineers is how to produce a HCI that allows for ease of use comparable to the old 'hard-desk', whilst exploiting the benefits of the extra information capability of such a software based system. The design is also often constrained by the project budget, both in terms of the hardware that can be purchased and the amount of time that can be spent on designing and implementing the soft screens. This guide seeks to give advice to ensure that the correct amount of hardware is procured and the correct level of software sophistication is achieved for any plant, without compromising operability and safety due to cost constraints.

In this guide the term 'Human Computer Interface' refers specifically to the hardware and physical screens that make up the operators desk, as well as the underlying software that constitutes and produces the visual images that appear on the screens - allowing information display and control of plant items. It does therefore include all the features that allow the operators to communicate with the software using Windows/Icons/Menus/Pointers (often referred to as WIMPS).

1.2 The role of the operator

The operator in a modern process plant has a number of different activities to perform. At an overview level these consist of such as, optimisation of production, ensuring compliance with environmental constraints, managing plant difficulties, co-ordination of maintenance etc. At the detail level the operator is concerned with actual operation of individual plant items, either at single component level (e.g. pump, motor) or a composite plant item level. In addition the actual tasks performed will depend on the current state of the plant - e.g. start-up, normal operation, change of mode of operation, abnormal situation, normal or emergency shutdown etc.

Whilst not necessarily having a primary objective of altering the operators role, most control desk refurbishment is often part of a larger control & instrumentation refurbishment which inevitably includes increased automation. Allied to the fact that a soft-desk is not designed for operating the plant by *continuous* manual control of individual basic plant items (e.g. opening /closing valves, starting/ stopping pieces of equipment), a redefinition of the operators role becomes necessary.

Typically then the role becomes much more one of a supervisor and monitor. Rather than performing the majority of control actions, the automatic system is allowed to perform that role, whilst the operator concentrates much more on ensuring that higher goals, such as maximising efficiency or limiting environmental emissions are achieved. In addition more time is devoted to analysing potential problems or dealing with actual shortfalls in performance. The operator tends therefore to have a more reactive than proactive role in the operation of the plant. Only when the plant moves away from its normal operating parameters, is major intervention, by the operator, required.

Therefore a modern HCI interface is required to perform satisfactorily for two very different situations. For most of the time the plant will be operating normally and the HCI must be designed to aid the operator maximise plant efficiency but when an abnormal period of operation occurs, the HCI must aid the operator in returning the plant to normal operation as soon as possible. Data that is vital to the operator for the efficient running of a plant may be totally useless in an abnormal situation and may in fact hinder the operator's attempts to rectify the problems, by hiding data useful to the rectification of the abnormal situation.

The HCI must then support the operator in all the different activities and scenarios with which he might be confronted. The design of the display formats, the navigation around the system and the actual physical number of screens become critical parameters in the design of a system which is crucial to the operators role.

1.3 Key design principles

The primary function of the HCI is to

Present the operator with a consistent, easy to use interface that provides monitoring and control functionality under all plant conditions.

The following categories of operation, in order of importance, need to be considered :

- | | |
|------------|--|
| Category 1 | <i>Abnormal Situation handling – This includes Start-up and Shutdown.</i> |
| Category 2 | <i>Normal Operation - Most utilised role</i> |
| Category 3 | <i>Optimisation</i> |
| Category 4 | <i>General information retrieval –This may be delegated to an IT system.</i> |

In order to achieve the above, many issues must be taken into account, ranging from the detailed design of individual display formats and the way that the display formats fit together in the hierarchy, through to the actual desk layout, number of screens and including the overall operational environment.

Whilst the purpose of the HCI is to provide a means to control a technical process, the design of the interface itself is very much a Human Factors issue. This interface is the nerve centre of the operator's work role. It is often the sole means that the operator has to obtain

data/information on the state and performance of the plant, as well as the means by which the operator can influence plant behaviour. In order to design the HCI it is imperative that the operator's activities are well understood - all the differing operational circumstances must be considered. An operator does not just have to cope with normal or steady state conditions, but with start-ups and shut-downs, plant condition changes and abnormal situations.

Whenever possible therefore :

A Task Analysis should be performed to capture the full remit of the operator's role.

Whilst the above will capture the operator's activities, a crucial element in the successfulness of the final interface is the acceptability by the operators. This, combined with the intimate operating knowledge possessed by most operators' means that all HCI projects should:

Mandate complete end-user involvement in both detailed display format design and the overall system design.

As emphasised above the analysis of activities will feed into the design of the HCI system. In converting this into actual hardware the following principle must be complied with:

The number of physical screens provided in the HCI must allow for complete access to all necessary information and controls under all operational circumstances.

One of the most important factors in operating a plant both efficiently and safely is being able to have an understanding of the state of the whole plant. On hardwired systems this was provided for by its very nature. All indications were on view, information access was simply a matter of scanning the panels. In a soft-desk implementation this is often forgotten or just not provided, but is an essential. Therefore:

The HCI design should allow for a permanently viewable plant overview display format.

Modern HCI, which are fully integrated will, almost without exception, also provide the alarm system interface. This is an important function in any process and due consideration should be given to the management and presentation of alarms. In this case:

Continuous access to alarm indication should be provided.

The reliability of the HCI must be considered - this must be appropriate for the type of plant under control i.e. how much failure is acceptable, what is the bare minimum of equipment that is necessary for running the plant.

Control rooms, processes and their requirements invariably change over time. The trend is for more information to be required, rather than less. Whilst it may be possible to build this into an existing hierarchy of display formats, it is inevitable that at some point, in order to maintain the operational functionality of the system that extra screens will be needed:

The capability to expand the number of physical screens should be built into the original design.

It should be remembered that if a HCI is to perform to its requirements and thus be accepted and exploited by its users (the operators) then ensuring that the Human Factors issues are addressed is as much a key factor as the engineering.

2. HCI design.

Section overview

This section provides the main content of the document. It identifies the major design issues that have to be taken into account when designing a HCI. It also indicates the main factors that influence the physical layout, the number of screens to be provided and hence the navigational issues that need to be considered.

2.1 General considerations

As noted in the introductory section the traditional hard-wired desk has evolved over a number of years and as with most products with this level of maturity, provides an excellent, ergonomically well designed and easy to use interface. In providing a soft-desk for the main operator interface, it is important not to lose the experience that has been built up from the traditional hard-desk installations.

A soft-desk HCI system may be introduced to a site for a number of reasons :

- Green field, new build site
- Complete replacement of an existing hard-desk / control system
- An extension to an existing hard-desk
- An upgrade to an earlier generation soft desk

The latter three often occurring due to a change in plant operating controls or the addition of new plant.

The issues to be addressed are similar in all circumstances, however the replacement situation has the additional problem of managing the large cultural change that such a project involves, whilst the extension situation brings it's own problems as two separate operating philosophies have to be integrated.

Experience has shown that the third situation should be avoided if possible.

2.1.1 Key items

In considering a soft-desk HCI installation a number of key items, which are almost taken for granted on hard-desks, have to be considered:

Information display - a standard back-panel has much to commend it in terms of basic information display. The meters and dials in their fixed positions are readily accessible, give excellent spatial awareness and provide overview and detailed data. Add to this the familiarity of the layout achieved by daily operation and the whole system becomes 'second nature' to the operator. Achieving the same level of awareness on a soft-desk requires careful thought and design and is discussed in detail in section 2.2.4 below.

Operational ease - A hard-desk provides an ergonomically excellent platform for taking control action. Pushing buttons and turning knobs are well-practised and familiar operations. With the fixed positioning of these controls, operators are able to quickly and easily manipulate them, sometimes in parallel. For example, on a conventional power plant with a number of pulverising coal mills, if there is a problem, the operator using a hard-desk, physically moves to the controls dedicated to the offending mill. On a soft-desk, he has to select the correct display format for the mill as well as focussing on the problem in hand. This takes time and mental effort. The dedicated positions of the hard-desk can provide a more direct mental link.

The methods of operator interaction for operating controls on a soft-desk need to be taken in to account as part of the overall design. In the past many HCI systems have been installed with the aim of mimicking the hardwired panel layouts that they replaced. This entailed the extensive use of Faceplate group displays and gave the impression of viewing the old hardwired panels through a window, a section at a time. For the operator, apart from

allowing full control from the desk, this virtual panel had few advantages. In fact the operator had gained a better "feel" for the condition of the plant by patrolling the old panels.

The objective of today's HCI designer should not be to mimic the panels of the past but to better them by using the strengths of the new technologies in the way data can be analysed and information presented.

Operator awareness - Soft-desk refurbishments are often 'part and parcel' of a full control system refurbishment, or in the case of new plant the soft-desk is the end point of a highly automated system. Whilst this now reduces the amount of low level action that the operator has to take, it is still important for awareness of the plant state to be maintained. On any system an operator is more likely to monitor variables if they are on display, rather than having to 'search them out'. On soft-desks the amount of 'screen real estate' is limited. Hence access needs to be as simple as possible (see section 2.2.4 below). An automated way of helping this can be to use a 'surveillance mode', whereby the operator selects specific displays and time intervals and the system cycles through these, displaying them on a chosen surveillance window.

Alarm provision - Increased automation provides a relatively calm operating scenario when plant is in a steady state. However given the restricted view of plant data on soft-desks (noted above) and the importance of alarms in times of upset the display of alarm information has to be given a high priority. More information on alarms systems can be found in (ref. 3)

The above general considerations raise a number of issues that have to be considered when designing a HCI for a process plant. These include the basic decisions that need to be taken when specifying the HCI and are described below.

2.2 Design issues

2.2.1 Physical layout and number of screens

Perhaps the most difficult consideration when designing a new HCI is the type and number of physical screens that are to be deployed. This can affect a number of the other considerations, e.g. navigational requirements, and in its turn is greatly affected by the influencing factors described in section 2.3. With the ever-increasing pressure on project costs, the number of screens to be used can easily become the focal point for cost reduction without due regard being taken to the operational consequences of the decisions taken.

The number of screens to be deployed should always be determined by the operational requirements of the system – including abnormal situations.

However any new HCI should be designed to allow additional screens to be added if, in the light of operational experience this proves to be necessary.

The optimal method for determining the number of screens is to perform a task analysis (ref. 6) of the operator's duties and then map the information requirements for each specific operational scenario onto screen availability. However this method is often prohibitively expensive and in the case of new plant it may not be possible to be able to fully determine all operational requirements until the plant has been commissioned.

It should be remembered that the number of screens deployed has a high impact on both the design of individual display formats and the use of the navigation system:

- A small number of screens can lead to a desire to overpopulate the amount of information on any given single screen.
- A small number of screens can lead to vital information becoming hidden from view, particularly during abnormal situations.
- A small number of screens can increase the amount of action required to navigate through the system in order to retrieve data and hence make the system less useable.

A case is often made for reducing the number of screens available, by stating that an operator can only deal with one or two screens at a time. Whilst this may be correct with regard to actively managing the screens, it is undoubtedly true that the operator can cope with large amounts of information on permanent view (c.f. the hard-desk back-panel). Often this will save time with regard to display format navigation, as well as potentially allowing a better mental context of the plant to be kept at all times.

It is also important to consider the amount of concurrent data that the operator may need to view.

It is totally undesirable to have a situation where the operator has to constantly change between two (or more) display formats to enable a task to be accomplished. This situation can arise when the number of screens has been underestimated. This then often leads to the situation where more information than is acceptable is placed on a single display format to try and remove the need for the display format switching.

A number of general points can be made with regard to determining the number of screens required for a given installation.

Determination of number of screens

- How many overview displays are required?
- What alarm display philosophy is to be taken?
- How much simultaneous control action may need to be taken?
- How much general monitoring is required?
- What level of multiple windowing is to be allowed?
- What level of redundancy is considered adequate?
- Can screens be used for more than one process, or 'module'?
- What future expansion may be required?

Fig 1 - Determination of number of screens required for a HCI

For control desk refurbishment one method of determining the screen requirements is to identify the main plant areas and by extracting operational experience from long standing operators work together to establish what overview information is required. Additionally an analysis of existing indications can be undertaken to ensure all-important parameters are covered. The operators can also be used to determine typical control scenarios for establishing control action requirements. This is of course a task analysis approach, but without perhaps the full rigour of a formal task analysis. Visiting other sites, both within the same industry and in external industries is a useful exercise in making operators aware of the possibilities that exist and in extracting operational experience.

This approach does of course require much co-operation and it has to be recognised that there can be a reluctance to express ideas or opinions during the design consultation phase. The use of prototypes can be a good way of iterating around the user's opinions. Experience has shown that often further iteration is required even after the 'final' system installation.

For example, the ability to cope with the following types of scenarios is required :

Actions to be carried out:

Steady state mode -monitoring, dealing with minor problems, increasing efficiency, environmental compliance.

Start up mode - -ability to display all plant items under consideration, plus overall process state.

Shut down -as for start up.

Abnormal Situations -sufficient screens to monitor all affected plant areas.

Additionally, the need for permanent displays, for overview information and the alarm system, should be factored into the calculation of the number of screens required. See also section 2.2.3 on navigation.

2.2.1.1 Use of Multi-windows

Most modern commercial systems allow for multi-windowing capability, however industry users are often divided about the potential benefits of this technology verses the perceived disadvantages. Ultimately, the extent to which multiple windows are used or restricted in a particular application is a decision for the system design engineer in consultation with the users. It should take into account the competency of the users (not only operators, but also supervisors, manufacturing engineers & technicians) and how comfortable they are with various styles of human interface.

The following table summarises some of the typical issues associated with the two types and will provide guidance to help decide which type of interface best suits a particular application.

Issue	Single Window Interface	Multiple Window Interface
User acceptability	Widely accepted	More likely to be accepted by operators who actively use desktop PC's running windows applications.
Ease of Use	Not normally an issue	May initially present a confusing picture to operators who are unfamiliar with windows. Typical problems that may be encountered include opening/minimising too many windows or losing windows off the edge of the screen. As a result, additional training requirements should be considered when installing or upgrading to a windows based interface.
Display Resolution	Not normally an issue	High resolution monitors are needed to ensure that graphics remain well defined and readable as they are scaled.
System Resources	Not normally an issue	Can be an issue if the workstation processor is incorrectly sized. Care should be taken when upgrading software on older systems to use windows and any additional memory to be installed should be sized conservatively.
View on the process	Depending on the number of screens available and the total number of graphic displays, a single window interface may restrict the operators view of the process and result in excessive paging between displays in order to effectively monitor process conditions.	Operators are able to set the windows up to suit a particular activity in hand and reduce the need for unnecessary paging. Engineers and technicians may find multiple windows extremely useful during trouble-shooting activities as they allow for a much larger range of process data to be monitored, again without the need for excessive paging. It is preferable for engineering workstations to be available for this functionality.
Complex Operations	Again, depending on the available screens, excessive paging may be required for complex operations such as plant start-ups.	With multiple windows, the operator can set up several windows ready to perform a complex operation such as a start-up. He can then quickly switch between windows at the appropriate time to perform each task rather than have to page through displays which can be time consuming and prone to error, particularly if operating "against the clock".

Fig 2 - Issues concerned with use of multi-windows

Depending on the competency levels of the users the following facilities may be made available:

- Allowing the user to easily resize the window, e.g. larger than standard size to make displays easier to read or smaller than standard size so that several displays can be monitored and controlled concurrently.
- Permitting different users to easily switch between single and multiple window modes (perhaps via an appropriate logon profile using preferences that the user can change or which are set up by the configuration engineer).
- Limit the maximum number of windows that can be opened to conserve system resources and prevent the possibility of confusion due to too many windows being active.
- Provide the ability to reset the view to a default setup or restore a previously saved set of windows and sizes/positions, e.g. to allow a supervisor or technician to continue an investigation or trouble-shooting activity.
- Allowing the configuration of important windows to always be on view

Furthermore, the use of a 'windows HCI' may have additional in-built features that could prove beneficial to some users e.g.

- Utilise user configurable quick-access buttons or "favourites".
- Provide a scratchpad window where notes or shift hand-over comments can be typed.
- Support a surveillance mode whereby the operator can configure a list of displays that will cycle through automatically.
- Provide a snapshot facility to allow the operator to temporarily capture a window so that the detail can be reviewed or discussed later.
- Give context sensitive help on all aspects on the human interface.
- Support the ability to access selected help documentation through a secure web based intranet, e.g. to quickly "hot link" to the relevant Standard Operating procedure (SOP) in response to a particular plant situation.

Multi-windows can be used, however any use must be carefully designed into the overall operator interface. Prudent use can enhance the interface, for example using windows to display small items of information related to a chosen hotspot. However free and indiscriminate use can negate a well-designed system.

2.2.1.2 Use of large screen displays

One issue, which now commonly occurs when designing a soft-desk configuration, is that of the use of large projection screens. These can provide a powerful addition to the operator interface, but have to be considered against the initial and ongoing running costs of such equipment. Additionally the mode of working within the control room will determine their suitability.

Advantages of using large screens are:

- They provide an excellent method for permanently displaying overview information and can replicate the old fixed back-panel of hard-desks. This will give a good spatial awareness capability.
- They provide a good display for occasions where a larger number of staff than usual are in the control room e.g. when commissioning the plant, if a problem is occurring or when staff who are not normally present in the control need access to information.
- On occasions when the control room is single staffed, they allow visibility of the process state from a distance, allowing the operator freedom to work at a different position within the control room.

- In control rooms where a supervisor is present, it is easier for the supervisor to keep track on the process without seeming to be interfering in an individual operator's work.

Appendix 2. gives examples of some configurations that have used large screen displays as part of the overall HCI.

2.2.2 Technology

The current large expanse of hardware and software technology available, often gives a plethora of options when designing display systems. The use of large screens and multi-windows noted above are the main issues.

The technology available also affects the choice of input device, which can now range from QWERTY keyboard, specialised keypad, mouse, tracker-ball, touch screen, light pen etc and can include facilities to allow tracking across screens by a single pointing device. Most systems are likely to use mouse input given the familiarity via the overwhelming acceptance of PCs elsewhere in the workplace.

Technology is also expanding away from the central control room with the advent of wireless handheld devices that can be used by attendants whilst out on plant to obtain graphical and mimic pictures of the process and other relevant information.

2.2.3 Navigation

The issue of navigation around a soft-desk is closely linked to the number of screens available.

For all but the smallest plant is impossible to permanently display even all the indications that were visible on an existing hard-desk. In addition soft-desk installations tend to have potentially many more parameters available for display. This is because either it is a new plant where the levels of instrumentation and processing tend to be much higher, or because advantage has been taken on a refurbishment to increase the level of instrumentation.

Navigational Requirements

The system needs to be :

- Flexible - to allow viewing of different data on different screens
- Easy to use - to allow quick and unambiguous selection of the required items
- Adaptable - to allow different methods to be used depending on individual preference within a well-defined framework.
- Responsive - to give quick, consistent and effective feedback to the user

Fig 3 - Navigational Requirements

Nearly all, navigational methods are based on a hierarchy of screens, which can either be deep or wide (or both!) and preferably with 'short cut' methods for instant access to known required display formats.

The essence of good navigation is the ability to retrieve the correct screen with the required information on it, easily and quickly, and to be able to 'find' the correct screen when the exact location of the information is not known.

Access to different display formats can be given in a number of ways:

Display format Access methods

- Via an index schematic, with a hotspot for each subsidiary display format
- Via a 'drop down' scrollable list of all screens - in a logical order (e.g. alphabetical, by plant area)
- Via functions keys on a hard (or soft) keyboard
- Via 'hotspots' (hot links) in each display format - giving access to a related display format
- Via 'hotspots' on actual plant mimics
- Via 'back', 'forward' and 'home' buttons
- Via access from other operational information e.g. from alarm lists, trends or database information.

Fig 4 - Display format access methods

The design of the display hierarchy can be crucial in enabling navigation throughout the existing display formats. It is usual to be able to move 'down' a hierarchy to obtain increasing levels of details about particular plant areas and plant items. Navigation 'across' the hierarchy from one display to another at the same level to allow access to display formats in related plant areas is also desirable. Access to control panels for plant item operation is usually at a fairly low level.

It is beneficial for access to be available directly from top-level display formats for important plant operations or for access to control panels, which cover a number of plant items (e.g. a full set of mills on a power plant). In this case, the display formats should contain all the detailed information necessary for control operations to be executed.

A simple level of hierarchy that is often used consists of total plant overview, 'unit' overview and then detail display formats.

It is important that access to display formats should not require multiple 'mouse clicks' or traversing the hierarchy from one display format to another. This can be time consuming and frustrating to the operator. Short cut direct access is therefore a very powerful aid.

Building up a plant hierarchy, which is logically connected, should allow sensible navigation to find the correct display format for any given plant parameter. As an operator aid, it should be possible to enter a full/partial tag number and be taken to the associated primary display.

Examples of display format hierarchies (for an oil refinery, power plant and chemical plant are given in **Appendix 3**)

2.2.4 Information access

A major difficulty with soft-desks is that the system contains much more information than it is physically possible to display at any one time. Data can be hidden from view and new ways of displaying information have to be developed.

An associated 'problem' is that of enhanced automation. Much plant operation is now automated with the operator acting in a supervisory role. This has led to suggestions that the operator should just be alerted to specific problems as and when the automation system detects them (the blank screen syndrome). This however is not viable. It prevents the operator from maintaining a mental model of the current plant-state. This model is invaluable for setting problems in context when they arise. The blank screen syndrome also prevents any pattern recognition being developed.

Thus, generally, the use of a blank screen syndrome approach is not to be recommended.

The main concept that should be addressed in information access is the use of overview displays. The HSE report into the incident at the Texaco Refinery, Milford Haven (ref. 4) singled out overview display formats as a major recommendation with regard to the Human Factors issues contributing to the incident:

"in a primarily display-screen operating system, the provision of good overview displays is of particular importance"

This advice has been borne out by experience in a number of soft-desk installations. Overview displays take on the role of the old hard-desk back-panel. They provide an 'at a glance' picture of the state of the main plant areas of a process and convey the essential information to determine whether any particular part of the plant needs attention. They prevent the situation where the operator becomes focussed on a narrow window of plant, ignoring or missing the potentially abnormal state that may be arising elsewhere in the process. For this reason it is recommended that

overview displays are kept permanently on display

In building up the set of display formats that are used in the display hierarchy, not only do the navigational aspects need to be taken into account, but also the types of display formats giving access to information. **It is best to produce a range of display formats dedicated to different aspects of information access.** These would typically take on such roles as:

- Plant area display formats
- Plant detail display formats
- Control panels
- Alarm list(s)
- Trend display
- Task orientated display
- Batch status
- Operator configurable displays
- Emergency Shut Down
- System/ network status

The main plant display formats will tend to follow the P&I Diagrams and be displayed as mimics. This is a powerful way of displaying the information, giving a good relationship with the physical plant and with operator understanding of the plant design. It can also be extremely useful in ensuring accuracy when communicating with staff located actually at a plant item, e.g. when carrying out maintenance.

Care has to be taken to ensure that P&I mimicking does not produce visual noise i.e. the amount of detail should be kept to a minimum.

Other information can be made available in high quality display formats, which may be best displayed in a variety of ways - bar charts, trends, etc.

Soft-desks also provide for much more detailed information access than is possible on a hard-desk - the full range of information in the plant database can be made available. This should be linked to individual items and an easy way of accessing this is via a pop-up panel (or faceplate), from the actual plant display format. Information relating to all aspects of the signal point can then be made available.

Ready made task orientated displays can also be developed. This enables the operator to have access to all parameters required for a specific task, available on a single display format. This replaces the need to have a number of different 'plant' display formats on display, in order to capture all the required information. The display can also help to guide the operator through the task steps.

These types of display formats help to reduce the number of physical screens required in a system.

2.2.5 Abnormal Situations Management

When plant deviates from the stable steady state running regime, swift action is often required. This highlights the need, emphasised above, for ease of use and quick access to the display and control capabilities of the system.

One approach is to create special 'emergency' screens (akin to the task orientated screens noted above) especially for known difficult situations. Of course many abnormal situations which occur cannot be foreseen. Here the creation of general-purpose display formats with key parameters that can be used in addition to the already noted overview display formats can be useful. None of these are foolproof, as the parameters required to be monitored in any incident will not be known in advance. The operation of any 'emergency' screen should also follow the actions used for normal screens, so that the element of familiarity is not lost, which is important in stressful conditions.

The potential use of the screens in an abnormal situation should be considered, to get an understanding of how many screens would be required to service this situation. The types of display formats needed and the concurrency required of them need to be assessed.

It is also prudent to consider a specific Emergency Shutdown (ESD) hard-desk. This would just contain the main plant indications and controls to shutdown the plant. It is also useful for guarding against a total failure of computer equipment.

Much abnormal situation management is concerned with the ability of the alarm system. This topic is fully covered in the EEMUA guide - "Alarm System - a guide to design, management & procurement" (ref. 3). It is recommended that the system alarm list be permanently on view. This would therefore account for the use of one screen in its own right.

A HCI is, potentially, a Programmable Electronic System under the IEC61508 standard (ref. 2). If it is part of a safety related system then

It should be designed, operated and maintained according to the IEC 61508 standard.

2.3 Factors influencing the overall design

The overall design of the soft-desk is governed by a number of influencing factors. There is no simple process that can take each of these factors and automatically produce an answer to the number of screens required, or the hierarchy of display formats. They have to be considered with regard to their overall influence on the system and judgements made as to how they affect the final outcome.

It is important to recognise that the design of the HCI cannot be divorced from the overall control room design and indeed must be seen in the context of meeting the overall control room operational and manning requirements.

The guidance aims to highlight the individual factors to be considered in each of these areas:

2.3.1 Automation

The amount of automation present on the plant is a crucial factor affecting the HCI design. Plants which are not highly automated, will require the operator to make many actions to operate the individual plant items comprising the plant - right down to opening/ closing valves, turning pumps on/ off etc. In dealing with a single process or sequence it may be necessary to have access to a number of these concurrently and hence the number of screens required is likely to be on the high side to ensure operability.

The other consequence of a non-automated plant is that the design of the display formats will be geared towards the operation of the individual plant items, rather than for more general overall control.

2.3.2 Plant size

The size of the plant to be controlled will affect the amount of data input to the HCI and the number of actions initiated from it. In general the larger the plant the greater the number of inputs to and outputs from the HCI. This will affect the amount of data to be displayed and hence the number of screens required and the number of display formats within the hierarchy. For a small plant it will be easy to use very few screens or display formats as the amount of data will be limited - this is also, of course related to the amount of instrumentation installed.

2.3.3 Process complexity

Complex plants tend to require a greater number of screens and display formats. It is more likely that a number of display formats will need to be on display at the same time to allow the operator a full picture of the state of the process. In addition these complex plants will need to display much detailed information. A balance has to be struck between the number of display formats that are designed and increasing the complexity of a smaller number of display formats.

2.3.4 Manning levels & multi-unit operation

The requirements on manning levels will affect the number of screens to be deployed. Obviously if a single unit is to have more than one operator, sufficient screens must be provided to enable each operator to carry out his required tasks. The system must also protect against confusion over 'who is doing what'.

However currently the trend is for at most a single operator (especially when plant is in a steady state). Some plants are now turning to look at a single operator for more than one unit. Problems presented here are:

- What is the minimum number of screens required for each unit - does this apply in all circumstances ?
- What configuration is permissible for sharing screens between unit - what minimum number should then be available to each unit ?
- How are displays designed to ensure the operator does not confuse which unit is being worked on ?
- What is the minimum acceptable configuration in times of system failure ?

The provision of 'read-only' screens can be useful for allowing access to other units or for use by different classes of staff.

2.3.5 Reliability/Redundancy/System Failure

The requirements for reliability and redundancy will affect the number of screens required and the configuration of such screens.

If the system is to be a full soft-desk with no backup hard-panel at all then the required reliability will be dependent on the required availability of the plant.

The failure requirements will be used as an input in determining the number of screens. No single failure of a component within the HCI should cause an overall failure of the HCI. Criteria should be specified with regard to the minimum number of screens to be available given a failure. This will affect the physical hardware configuration of systems - for example how many screens should be driven from a single workstation, can screens be allowed to be used for different units/ processes in times of failure ?

It is expected that most HCI's will be designed with a high level of redundancy in terms of duplication of, screens; communication networks; power supplies (mains & UPS) etc

However, the possibility of various types of HCI failure still need to be considered in terms of defining specific operator responses to cover all possible eventualities. One way of achieving this is to include a section on HCI failure within the plant Standard Operation Procedure (SOP) which would typically cover the following scenarios:

- Loss of mains power and steps that need to be taken before the UPS batteries expire (assuming that there is no generator backup).
- Total failure of a single screen and procedures for contacting maintenance personnel to trouble-shoot and fix.
- Total failure of all screens for a plant area. Here depending on the stability of the process prior to the fault and whether an independent trip system exists, it may be possible to continue to operate the plant for a short period of time to allow the fault to be investigated before initiating a shutdown. If the process was unstable prior to the fault then it would be probable that a manual shutdown would be warranted.
- Other hardware failure e.g.
 - A single screen or multiple screens continue to work but stop updating.
 - A single screen or multiple screens become sluggish.
 - Alarms stop reporting to a single screen or multiple screens.
 - Historic trends stop updating.
 - Failure of a peripheral e.g. a keyboard, trackball, or printer.
 - Response to system fault alarms either wired to the DCS or generated internally.
 - Other types of failure not covered elsewhere.

Whilst it is financially desirable to produce least-cost solutions to the HCI in terms of minimising the amount of hardware, this must not be done at the expense of compromising operational and safety needs.

3. Display format design guidance

Section overview

This section provides an introduction to display format design issues.

The design of individual display formats will depend very much on the overall hierarchy and navigational decisions taken. These will govern the types of display formats, the level of detail needed and the way that extra functionality is provided on each display format.

Many companies have internal guidelines on specific display format design, in terms of colour, fonts, animation etc and indeed international standards are also in preparation (ref. 5). This section only presents some of the higher level considerations and practice that should be debated when designing display formats. It is not intended here to give detailed design guidance on display format design. The ASM consortium, have produced excellent guidelines on this subject (ref. 1) and these should be consulted for a thorough coverage of the subject

3.1 Principles of good display & interface design

Some general principles can be stated:

Clarity - it should be self-evident what the purpose of the display format is. Individual items on the display format should be obvious. Graphics, text and numeric displays should be clear, easy to read and it should be obvious to what they refer. Cool and subdued colours should be used in normal operation, to allow bright colours which will stand out, to be used in abnormal situations.

Consistency - all graphic indicators, text and colour coding should be consistent, not only within a display format, but between display formats in the hierarchy. The operation of a control (e.g. hotspot/ hot link) should have the same effect on all display formats.

Variety - a limited variety of display techniques should be used, so that operator's can become easily familiar with them.

Feedback - any control or other action on a display format should give feedback to the operator to give awareness that the system is performing the requested task or indeed is giving an error. The feedback needs to be near instantaneous. Feedback on completion of the action may also be required (or on failure of the action).

Robustness - where interface action is taken by the operator the system should be designed so that it can cope with incorrect key strokes/ mouse clicks and so that the operator can return to the original position if need be.

Failure - failure of a display or of items on the display should be immediately apparent to the operator.

Redundancy - multiple information display should be avoided, unless this is required to achieve specific reliability requirements or as determined by task analysis.

Demand v Status - indications should make clear what values are indicating actual plant status and what are indicating set-points or demanded values.

Spatial Variation - if situations exist where several displays depict similar process units, then it can be beneficial to place similar objects in different positions, to help differentiation between the units. Note the clash with the concept of consistency.

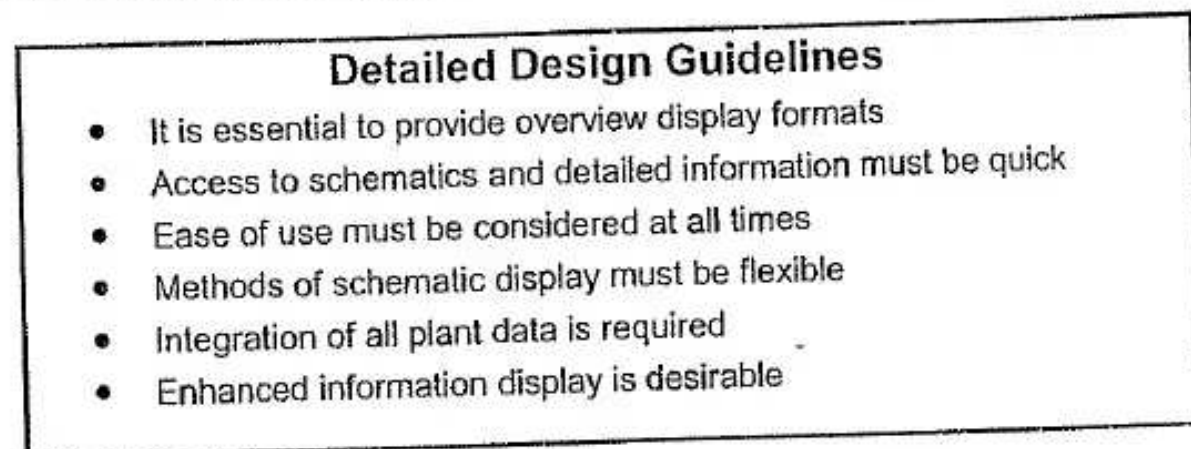


Fig 5 - Detailed design guidelines

3.2 Display format design methodology

For each display format that is to be designed the main factor is a proper and systematic analysis of the proposed use of the display. This is true whether the display format is to be basically plant based or whether it is to be task based.

The designer must consider for each item of information:

- Whether quantitative or qualitative information is required
- Whether comparisons with other data is required
- The accuracy with which the data must be read
- The degree of detail that is required
- The characteristics of the data - rate of change, range etc

3.2.1 Display format configuration & layout

The overall look and feel of any HCI system should be consistent. A 'house' style should be developed and documented and all display formats should follow this. It is important to ensure that the overall design of the display formats should be well thought through and promoting the type of display formats where it is easy to assimilate the information displayed.

Information density:

- Display packing density should be less than 50% and preferably around the 25-30% mark. (i.e. less than 50% of available area should be covered with text/graphics)
- Display screens should not appear cluttered. Grouping data and placing it on backgrounds can help to avoid a cluttered appearance.
- Spatial alignment helps to give a good appearance.
- It should not be necessary to have more than one display on view when performing a specific task

Data organisation:

- Information should be grouped in order of frequency of use, otherwise information should be functionally grouped
- Grouped data should be arranged with consistent placement of items so that the user is aware of changes in the data - e.g. difference, trends, similarities.
- Other data should be grouped logically e.g. alphabetically, numerically etc.
- When grouping the same items on different display formats, they should, if at all possible, be grouped in the same position and in the same layout, to facilitate pattern

recognition. However if display formats are the same but just referring to a different instance of a particular plant item (e.g. mills on a power station unit, tanks on an oil refinery), great care needs to be taken to ensure confusion does not arise. In this case the spatial variation referred to above can be useful.

Frame layout:

- Header information should follow a consistent layout and style.
- Display format names should be consistent with their content.

3.2.2 Information display

The general principles noted above also apply here.

Information display

Information can be displayed in a number of ways - these are basically:

- Tabular
Data lists

These should only be used as part of detailed displays, where access to precise information is required. They do not provide for easy, quick access to information. If used a number of points can be followed to ease access - e.g. one item to a line, enumerate each line, list common options first, display headings.

- Graphical
Analogue (dials)
Digital (counters, numerical)
Binary Indicators
Bar Charts
Trend plots
Symbols (valves, pumps)
Vessel fills

This refers to individual items that can be used as parts of a mimic display. For each type of item a set of standards should be developed, dealing with the detailed design of the item. For instance numbering should increase clockwise, left to right or bottom to top, numbers should be outside graduation marks, zero and maximum values on circular scales should be at 7 and 5 o'clock.

- Mimics
Mimics should have a direct reference to the plant, and give an overview of a plant system. Both abstract and pictorial 'shapes' can be used as appropriate. The minimum amount of detail required to give a meaningful representation should be used.

Guidelines for each of the above should be developed.

Visual coding dimensions

Additionally, the following items are the commonplace building blocks for developing display formats. Again guidelines need to be produced to ensure consistent use of each of the factors.

- Text
Text can be used in a number of ways - Messages (to convey status/ information/ diagnostics), labels (for identification), data (i.e. alphanumeric indication), or as an abbreviation.

- Shape
The shape of items can be either Abstract (which has to be learned before it can be used) or Pictorial (which references the real world and therefore is easily understood)
- Size
This is applied to both text and shape.
- Colour
The use of colour can add significantly to a display, but should not dominate. Indeed it is a useful exercise to design for monochrome. Care needs to be taken - display formats are often designed without any real thought to the outcome and many garish and 'loud' display formats exist which merely mitigate against their desired use. A cool background colour should be used.

Visual Enhancement:

The basic coding facilities are often enhanced by more sophisticated techniques. Unfortunately these are often used to create displays which may be visually exciting, but not necessarily fit for their purpose. The two main techniques are animation and flashing:

Animation –

Animated graphics should only be used sparingly and where absolutely necessarily to highlight a significant change. For instance using animation to show rotating plant or flow in a pipe adds very little to the display format, can be distracting and will more likely than not add significantly to the processor load. Therefore it should generally be avoided.

Flashing - this technique is excellent for attracting attention, but if overused or used without thought, can create an effect where it is difficult to read the information or difficult to distinguish the message that is trying to be conveyed. This is especially true if flashing more than one item when the flash should be synchronised to avoid even more distraction.

Level of resolution

Digital indicators should only show information to a level of resolution appropriate to the use of the display format and to the accuracy of the signal. For instance on high level overviews it is unlikely that resolution would be needed to a single decimal point for signals whose normal range is in the hundreds. On a detailed display, used for analysing an error situation, then a high level of resolution may be appropriate, but the resolution of each signal should be considered on its own merits. Due regard must also be taken to the liveliness of the signal being displayed, as displaying high resolution is not useful if the signal is very active

4. Control room design

Section overview

This section provides an introduction to the overall control room design.

This guide is not intended to address the detailed issues regarding the overall control room design. Many factors have to be taken in to account here and there is much in the literature about control room design. Additionally many Human Factors consultancies exist with a wealth of experience. The use of a specialist control building architect will produce a better designed building, which will often also be smaller and cheaper, both initially and over the life of the building. This section is therefore just intended to provide a high level overview of the issues that need to be addressed when designing a control room.

4.1 Factors influencing

The following is intended as a short checklist of the factors that need to be considered in the overall design of a control room :

- Different users of the control room
(Operators, maintenance staff, computer staff, managers, cleaners, visitors etc) - these should be divided into primary and secondary users. The differing needs and requirements of each set of users need to be considered to ensure that there is no conflict and to enable each user to perform their role to the benefit of all.
- Control room access levels --
The means of access by the different users and the areas within their scope need to be clearly defined, to prevent clashes of interest.
- Environment
 - Blast resistance
 - Lighting
 - Heating & Ventilation
 - Noise levels
 - Furnishings
 - Colour schemes

The overall environment can contribute heavily to the effectiveness of the control room staff and indeed affect the perception of the HCI. Detailed information on the areas listed is widely available

- Console design
 - Console & related furniture
 - Size of monitors
 - Adjustability of keyboards/monitors
 - Writing space
 - Power supply, cabling
 - Phones
 - CCTV monitors
 - Radios
 - Printers
 - Storage requirements
 - Hard wired ESD panel

Again, the physical layout and furniture used for mounting the HCI equipment can have a large bearing on the acceptability of the total product. Built in 'television' studio type panels are impressive, but the current trend is to move from these towards more 'desk top' configurations. These can be very effective, but have to be carefully designed to ensure that the total image is not one of clutter or haphazardly placed monitors.

- Safety requirements
 - HSE regulations
 - Fire prevention
 - Fire control
 - Emergency exits
- Other Operations support requirements
 - Meeting room
 - Office facilities (photocopier)
 - PCs (if not incorporated into the console).

Whilst not directly affecting the operator interface, provision of a well placed and functional mess-room is an essential part of the control room. It should especially be borne in mind that good operation on an HCI is primarily a sign of good Human Factors design and the mess-room can have a significant affect on operator performance.

It can be useful for the operator's mess facilities to provide viewing access (one way) to the control room, so that they can still be aware of any disturbances.

- Engineering Facilities

In large control rooms, it can be beneficial to install additional segregated workstation bays for use by supervisors or engineering personnel. These can be used for tuning, handling upsets or system configuration. In some plants these are installed in separate engineering rooms or offices.

5. Overall HCI system design guide

Section overview

The previous sections of the guide have presented the issues and factors affecting the design of a HCI for process control systems. They have been intended to give a good grounding in what needs to be considered when approaching a HCI design. In addition some solutions and techniques have been suggested. The following sub-sections are intended to bring all this information together and provide a methodical approach to the design.

5.1 Requirements

The most important aspect of designing a HCI system is to understand, as fully as possible, what tasks it will be expected to perform and how it will be required to perform them. The forms in appendix 5 or a modified version of them may assist in this.

It is important to separate the essential requirements from the 'nice to have'. A HCI system will normally be expected to fulfil a number of roles as discussed in the previous sections. The designers must clearly understand the priority of each of these roles and should ensure that the methods used to achieve one particular role will not reduce the effectiveness of the system in achieving a higher priority role. This is particularly important when using multi-windowing techniques and importing data / information from non 'control network' sources.

Two opposing driving forces often dictate the size of a HCI system:

- The pressure to make it as small as possible to minimise costs and save space.

This can lead to over cluttered screen displays, making information difficult to read and understand. It may also require the use of more complex navigation techniques so that information can be retrieved easily. The reliability of the system can also be impacted.

- The temptation to add additional screens / workstations because the cost of them has fallen significantly in recent years.

This can cause problems similar to those encountered with DCS alarm proliferation. (i.e. the operator being presented with too much raw data because it is in-expensive to provide and because there is space available to display it).

It is suggested that the role of many HCI systems fall into some or all of the following priority groups.

Group 1 (Highest)	Abnormal Situation handling	This includes Start-up and Shutdown.
Group 2	Normal Operation -	Most utilised role
Group 3	Optimisation	
Group 4 (Lowest)	General information retrieval	This role may be delegated to an IT system.

5.2 Large Screens

The decision on whether or not to use large (project based) screens in the control room will be based upon a number of factors, the most import of these being (apart from 'is there space to fit them')

How do you operate normally and how do you handle a protracted abnormal situation? "

The purpose of the screens should also be clear - are they to assist the operator and his/her colleagues or are they for casual visitors to the control room.

Should people be encouraged to gather around the operator to discuss problems and offer advice?

Should the operator be left to deal with his immediate problems without distraction while other personnel review the overall causes of the problem in relative calm, possibly on a separate HCI read-only console or in an adjoining room equipped with a read-only HCI workstation and large screen?

In an abnormal situation a large screen will become a focal point for people to gather and discuss the situation, particularly within normal working hours.

It must be decided, whether this is an advantage or disadvantage for the type of operation under consideration.

Note - that in normal operation large screens can provide good overview display capability.

5.3 Number of Screens

There is no magic formula that can give an answer to 'how many screens' are required for all potential scenarios. Again the designers must understand, as fully as possible, all of the tasks that the HCI will be required to perform. The following methodology, using a step by step approach, should assist the designers in making the final choice.

1. Availability requirements

What are the minimum number of screens / workstations required to provide the level of availability needed.

If redundancy is a requirement then, even for a small HCI system, at least two workstations will be required. Each workstation would drive one or two screens.

2. Overview requirements

Overviews that provide the operator with a good situational awareness of the state of the Units/ equipment for which they are responsible, are invaluable. They assist in the prevention of 'tunnel vision' during plant upsets. Permanently displayed overviews should be considered for all continuously attended HCI systems.

3. Alarm requirements

A permanently displayed alarm page should be considered for all HCI systems, which are continuously attended.

4. Abnormal situation management

Having designed the ASM display formats (see display formats below) it should be possible to judge the minimum number of overview and control screens required to handle an abnormal situation. A task analysis should then be conducted to assess the performance of your chosen number of screens/workstations against a possible abnormal situation - one which will require the most operator intervention. At this point the possibility of one workstation failure may be considered.

This should be an iterative process until the design of the ASM display formats and the number of screens/workstations proposed, allow the

operator to handle the most onerous abnormal situation scenario effectively but with the minimum number of screens.

5. The type and number of tasks undertaken in normal operation (monitoring, controlling etc)

An analysis should now be carried out on tasks, which under normal operation require the most HCI activity. This will help to assess whether the size of the proposed system is sufficient to handle normal operation.

6. Optimisation and general information retrieval

Finally the need for additional screens / workstations to display economic and procedural information / data should be considered.

This information can often be displayed without additional hardware by the use of windowing or navigation techniques.

In an abnormal situation optimisation data is normally useless but procedural information (e.g. Operating procedures) may be invaluable - even so it must not reduce the operability of the control system.

7. Future Requirements

It is essential to allow for future expansion, as invariably operational procedures change over time. This should include an allowance for physical space.

Example 1.

A medium sized continuously attended HCI system is required for a new batch process plant. Redundancy is required so that a single failure does not result in the loss of the whole system.

It is decided that one screen can provide all of the overview information required and that one alarm screen will be provided.

The designer decides that the minimum hardware required is 2 workstations and four screens.

This will provide 1 permanent overview screen, 1 permanent alarm screen and 2 control screens.

An abnormal situation task analysis shows that the operator can easily handle the required workload using 2 control screens and the specially designed ASM graphics.

A normal operation task analysis also shows the proposed configuration to be adequate.

The plant operator uses a large amount of optimisation data and requires the use of the HCI to retrieve a number of procedures due to the variety of processes that the plant can be used for. It is decided to provide an additional workstation and screen to accommodate this.

The final HCI consists of 3 workstations and 5 screens.

Example 2.

A medium sized continuously attended HCI system is required for a new continuous process plant. Redundancy is required so that a single failure does not result in the loss of the whole system.

It is decided that two screens can provide all of the overview information required and that one alarm screen will be provided.

The designer decides that the minimum hardware required is 3 workstations and six screens.

This will provide 2 permanent overview screen, 1 permanent alarm screen and 3 control screens.

It is decided that the abnormal situation, which requires the most HCI activity, is a plant start-up situation. A task analysis shows that the operator could handle the required workload using 3 control screens and the specially designed ASM graphics.

It is noted that the failure of one workstation would leave two still working and taking into account the mean time to repair/replace the faulty workstation, this is considered acceptable.

A normal operation task analysis also shows the proposed configuration to be adequate.

The plant operator uses a small amount of optimisation data and retrieval of operating procedures etc. is via an IT workstation.

The final HCI consists of 3 workstations and 6 screens.

Some real examples of screen configurations are provided in appendix 2.

5.4 Navigational techniques

A number of navigational techniques exist, as described earlier. A well-designed system should seek to give choice in navigating the system whilst avoiding confusion. A consistent approach to each and every display format should be promoted, so that the same actions always have the same effect.

A single preferred technique should be promoted to all operators who are new to the system. This technique should be intuitive and built upon the alarm-display relationship. (i.e. an alarm shown on the overview, alarm screen or keyboard indicator should lead the inexperienced operator easily to the display format that he/she requires to action it). As the operator's experience with the system grows additional techniques can be employed, which aid workflow or provide shortcuts to specific display formats.

The use of the following techniques can be considered:

Overview / plant area / detail hierarchy using vertical and horizontal navigation

Links from plant objects

Access to control actions - pop up templates

Soft fast access 'keyboard' buttons

Hard fast access keyboard buttons

Index display format

Index list of display formats

Numbering/naming of display formats

Hotspot shortcut buttons to different (but associated) display formats

Back / forward / home buttons

Pop up lists / pull-down menus

Some examples of navigation techniques are shown in appendix 3.

5.5 Windowing

As described in section 2.2.1.1 multi-windowing can provide many benefits if used correctly. If used incorrectly it can cause confusion and mask important information at times of high stress.

If the main objective of the HCI is to control the process then multi-windowing should be restricted for use to such tasks as pop-up face-plates, trending or simple instructions.

The operator should not normally be allowed to resize or drag windows across a display screen. These facilities can cause problems, particularly in the hands of an inexperienced

operator. It is also impossible for the display designer to take account of such variations when constructing display formats etc.

Display formats are designed to be displayed at a certain size, reducing this size may make text difficult to read at normal distances and cause visual fatigue.

Dragging windows across the screen may disorientate the operator at moments of high stress.

The position on the screen of particular types of information / data should be consistent at all times.

Dragging one window across another may also mask important information.

Many manufacturers offer a range of other information displays via the HCI, for instance data sheets, maintenance sheets, manuals etc. Much of this functionality is more appropriate to the maintenance / engineering staff and only those functions that can directly help the operator should be made available on the operator HCI.

5.6 Format Considerations

The types and utilisation of individual formats play an integral part in the overall structure of the display hierarchy and how to navigate it. In designing the hierarchy the following types of format should be considered, along with their interactions:

Graphical overview display formats

These displays provide an easy to assimilate graphic, which depicts the condition of the plant for which the operator is responsible. It should be possible to invoke a particular ASM display or Control Display directly from an overview object / soft-key.

Alarm displays

Alarm lists/panels should be provided which allow filtering by priority or area and which if possible can invoke a particular ASM display or Control Display directly from the chosen alarm object.

Abnormal situation management (ASM) display formats

These displays should provide the operator with all of the control and monitoring functions required, to deal with an abnormal situation on a particular area or unit.

Control display formats

These displays should allow the operator to control and monitor specific plant and equipment in normal use. They may include P&ID or flow-chart type display formats.

Detailed display formats

Detailed display formats are generally used for task-orientated displays or close monitoring and control of particular parts of the plant. They are not normally used as often as control formats and are usually invoked from them, as a way of 'zooming in' on a control display format section.

System status display formats

These display formats should display the health of the HCI system as a whole and allow any faults on the system to be detected and located in the quickest possible time. A graphical representation of the system topology would make a good starting point with detailed display formats being called up to provide more information on specific areas of the system.

5.7 Maintenance Life cycle.

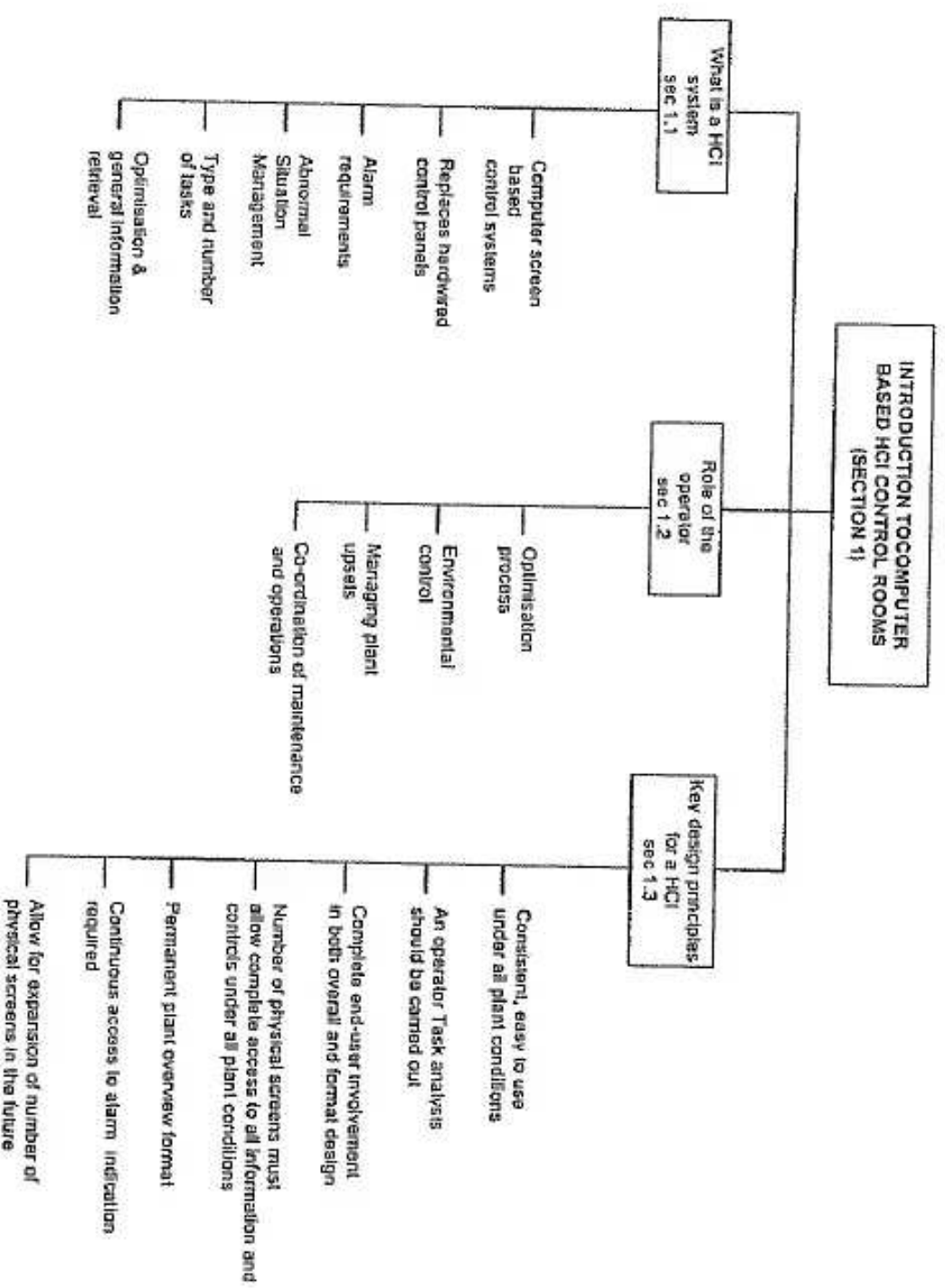
Once an HCI is in service, the usability (e.g. speed of response) of the system should be monitored at regular intervals (for instance bi-annually), following major modifications to the plant or system and after abnormal situation occurrences to determine if any improvements are required.

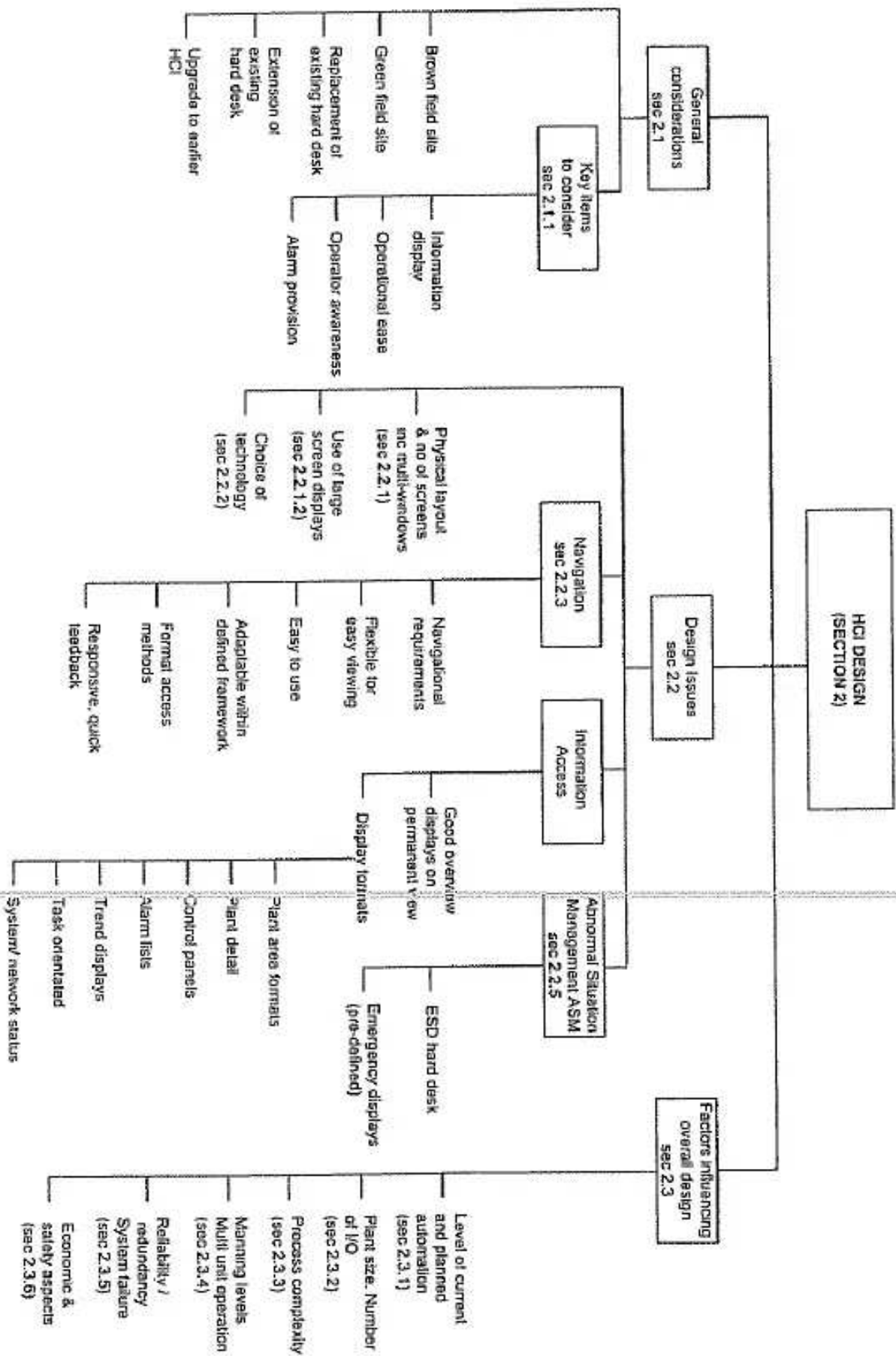
- The forms that are provided in appendix 5 and appendix 6, or similar ones should be used to assess this performance.
- Log files may be used to analyse operator activity on the HCI and help identify weaknesses in the navigation mechanism or display format design.
- Operator interviews will give the users perspective of how the HCI is performing and how it can be improved.

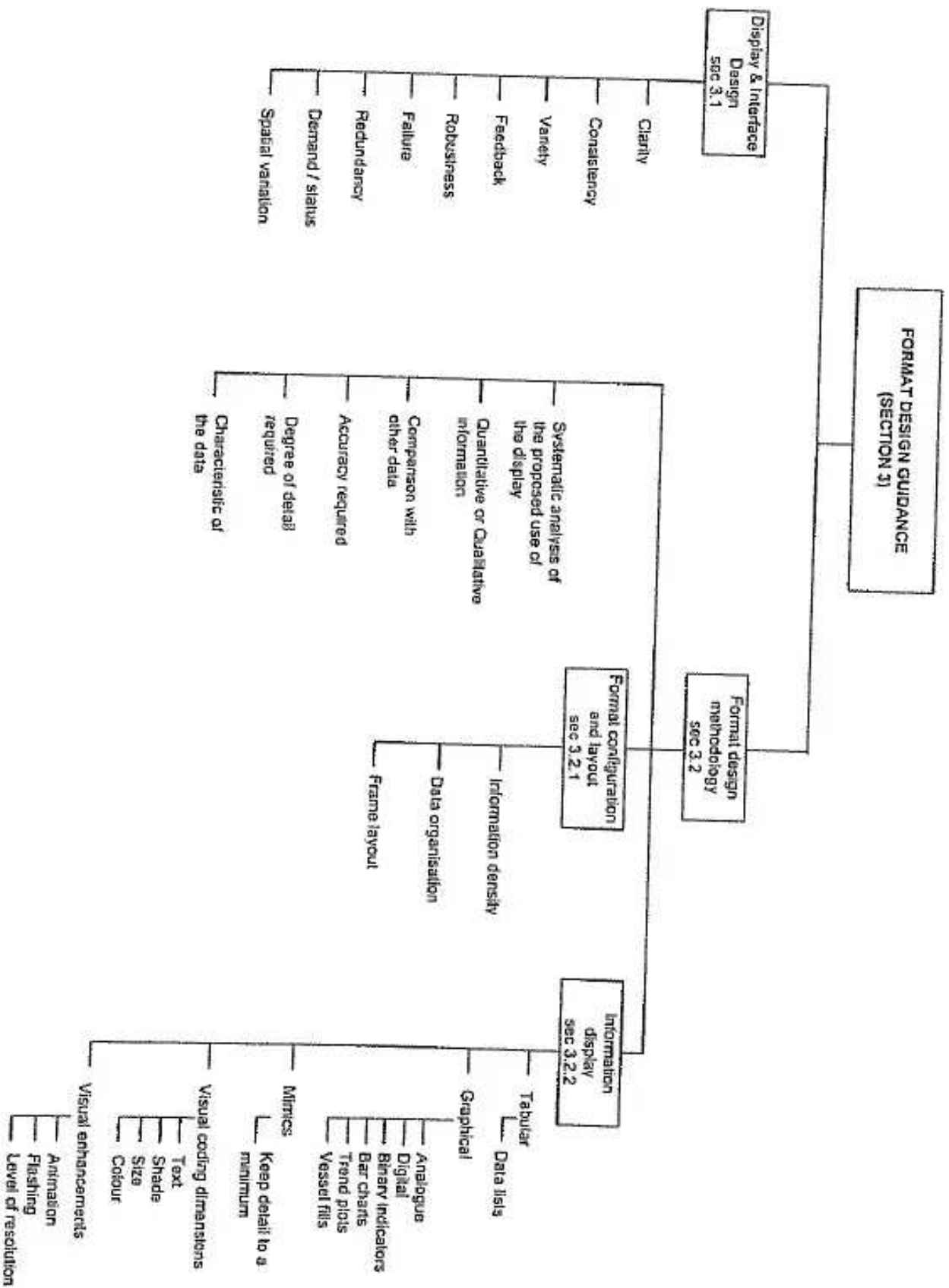
One of the advantages of modern HCI systems is that they may easily be modified in the light of experience. Modification should always be subject to change control procedures. It must be remembered that any improvements to one of the roles for which the HCI is used should not reduce the effectiveness of the system in a higher priority role.

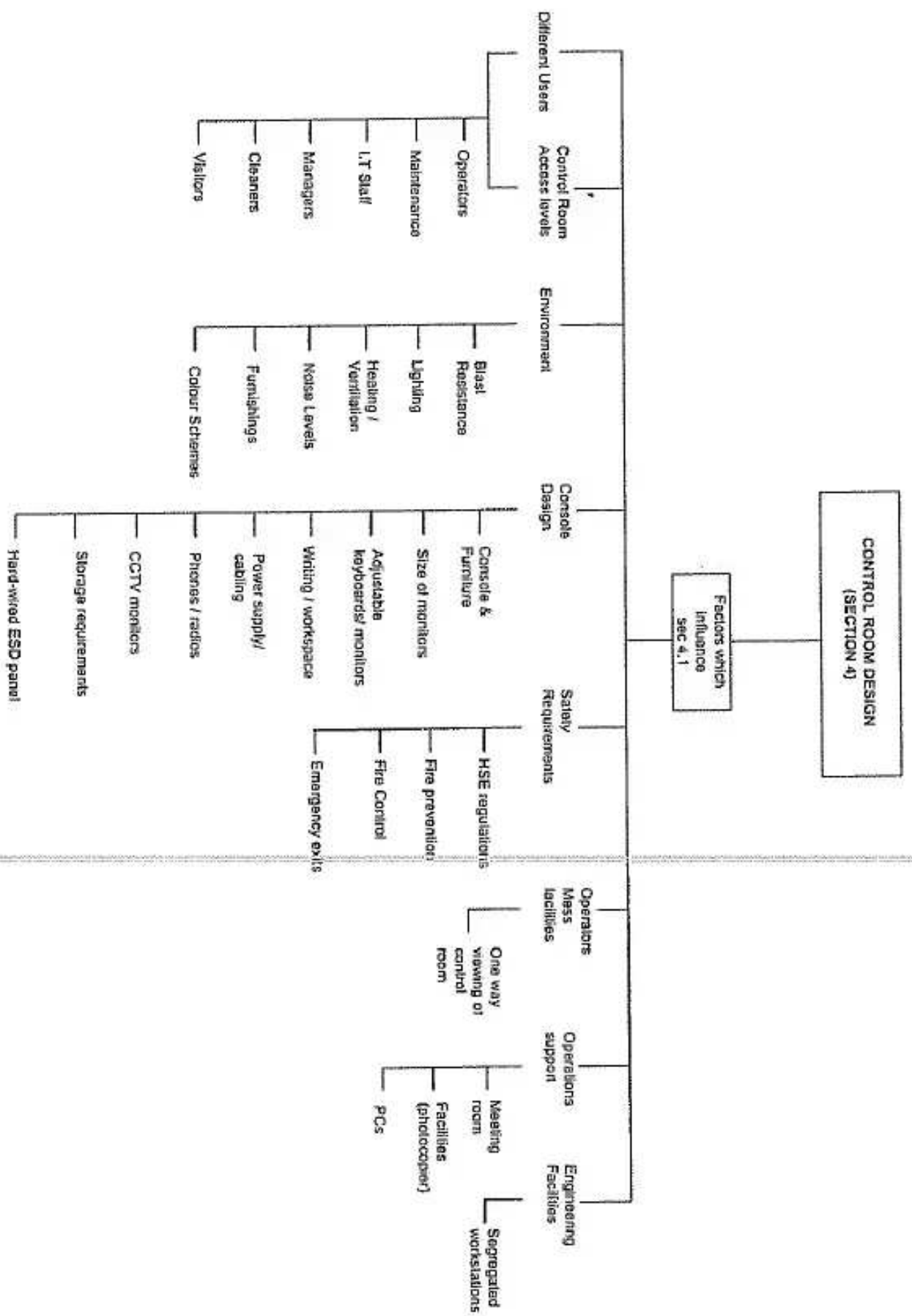
5.8 Guidance for a well designed HCI

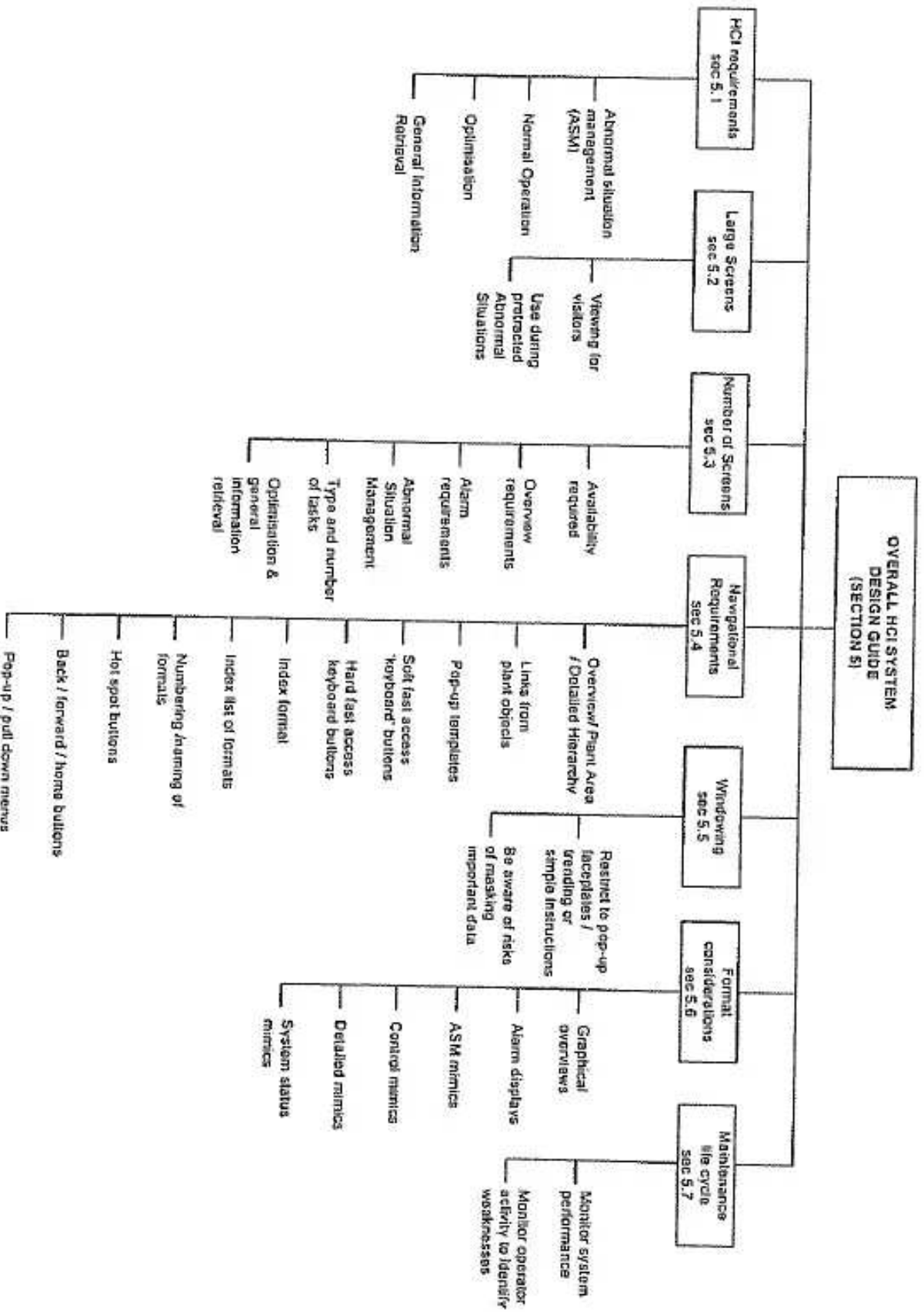
The following five pages set out in a graphical form the information contained in the earlier chapters of this document, as well as this chapter. They are meant to give a quick and easy reference to the main points contained in this document.











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Appendix 1.

Glossary

This appendix provides a glossary of some of the terms used in the Guide. It is not intended to be comprehensive, but to concentrate on those terms that are likely to be unclear or cause confusion.

Hard-desk	A hard-wired fixed position physical control desk, with meters and indicators for display purposes and knobs and push buttons for control operation.
Back-panel	A hard-wired panel, used for displaying indicators and meters, where little or no control action needs to be taken.
Soft-desk	A display and control desk, where all display and interaction is performed via computer screens and computer input devices.
Screen.	Actual physical device upon which a display format or schematic is displayed.
Display format	A software-generated display, usually, in the process industry, containing a representation of physical plant.
Schematic	Another term for display format. Sometimes used to describe a component of a display format.
DCS.	Distributed Control System.
SCADA	Supervisory Control And Data Acquisition.
ESD.	Emergency Shutdown System. An automatic protection system which will act to shut down the plant if it enters a potentially dangerous state. In some countries this is called a Safety Instrumented System (SIS).

Appendix 2.

Example screen configurations

This appendix provides examples taken from installations that have actually had their control desks refurbished in the last few years, or have been a new build installation.

Large coal fired generation plant:

(1) Didcot A (Innogy plc)

A 4x500MWs plant situated in Oxfordshire. This has been the subject of a rolling refurbishment programme over the last few years. The initial refurbishments, whilst providing a full soft-desk functionality (i.e. not only were displays present, but control functionality was included), did not remove the 'back-panels' so typical of power generation plants. The refurbishments have now moved on with two units being fully converted to soft-desk and the other units in the planning stage.

The final configuration consists of

- 3 x 51" x 38" projection screens - used to display three overview display formats (boiler, turbine and feed system)

- 6 x 40" x 30" monitors

 - One monitor is dedicated to the alarm list

 - One monitor is dedicated to the commercial load following display

 - Two monitors are used for control actions

 - Two monitors are used for general monitoring of the plant

The computer based system is fully dual redundant.

(2) Drax (AES Drax Power Ltd)

A 6x660MWs plant situated in West Yorkshire. The plant was built in two phases and the control desks have been refurbished on units 1, 2 and 3. This is a full soft-desk installation, with dual redundant computer based systems. Each unit has:

- 3 x 51" x 38" projection screens - used to display three overview display formats (boiler, turbine and feed system)

- 9 x 40"x30" monitors

 - One monitor is dedicated to the alarm list

 - One monitor is dedicated to the commercial load following display

 - One monitor is dedicated to the technical load control display/control

 - Three monitors are used for monitoring/ control actions

 - Three monitors are used for general monitoring of the plant. For instance these can show four trends each, to enable non-digital representations of plant dynamics.

In addition the operator's console has a CCTV, an operational information archive system screen and also an office PC.

Chemical Industry

Barry (Dow Corning)

- 6 x 21" monitors per operator

- 2 x 21" CCTV screens with remote control

- 1 x hardwired annunciator

- dedicated alarm beacons for Safety/ESD/Process alarms

*Oil refineries**(i) Lindsey Oil Refinery Ltd*

A large petro-chemical plant owned by TotalFinaElf and situated in North Lincolnshire. The Stage 3&4 control room instrumentation is being converted from panel based control to soft-desk using a recognised DCS vendor and with fully fault tolerant configuration. Three panel operators are located within the control room and are responsible for multiple process units.

The process is a continuous 24 hour 7 days a week operation.

The configuration for each operator is as shown in fig.1 of appendix 3 and consists of:

6 x 20" Flat screen technology monitors, double stacked.
3 x Operator Keyboards
3 x Pointing devices.

Two monitors are dedicated to monitoring graphical overviews (SVS) of the units.
One monitor is dedicated to Alarms.
Three monitors are dedicated to general control and monitoring.

(2) Coryton (Mobil-BP)

The following process units are used with displays as indicated:

Lubes Process Units:

10 screens	6 process plants	2 panel operators per shift
------------	------------------	-----------------------------

Product Movement Area

6 screens	5 Off-sites areas	1.5 panel operators per shift
-----------	-------------------	-------------------------------

Fuels Process Units

10 screens	7 process plants	2 panel operators per shift
------------	------------------	-----------------------------

Isomerisation

4 screens	1 process plants	1 panel operator per shift
-----------	------------------	----------------------------

Cracking Process Units

13 screens	10 process plants	2 panel operators per shift
------------	-------------------	-----------------------------

Utilities/Boiler Units

6 screens	4 process units	1 panel operator per shift
-----------	-----------------	----------------------------

Appendix 3.

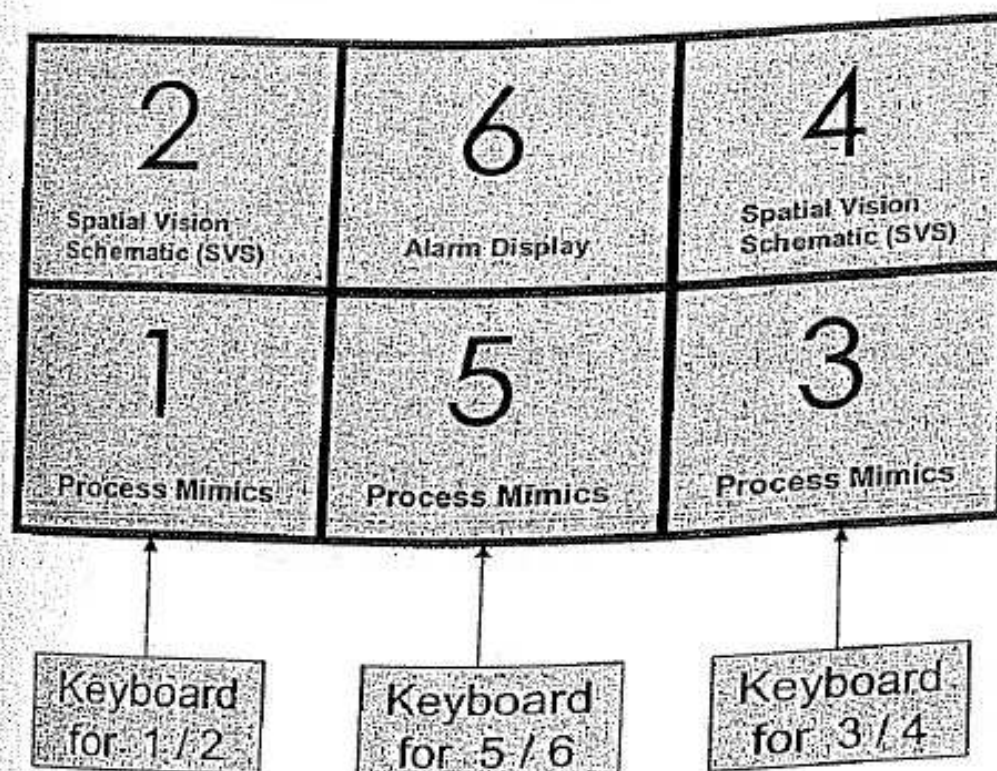
Example display hierarchies

This appendix provides examples of how display hierarchies can be constructed and the methods of navigation through them.

Lindsey Oil Refinery refurbishment

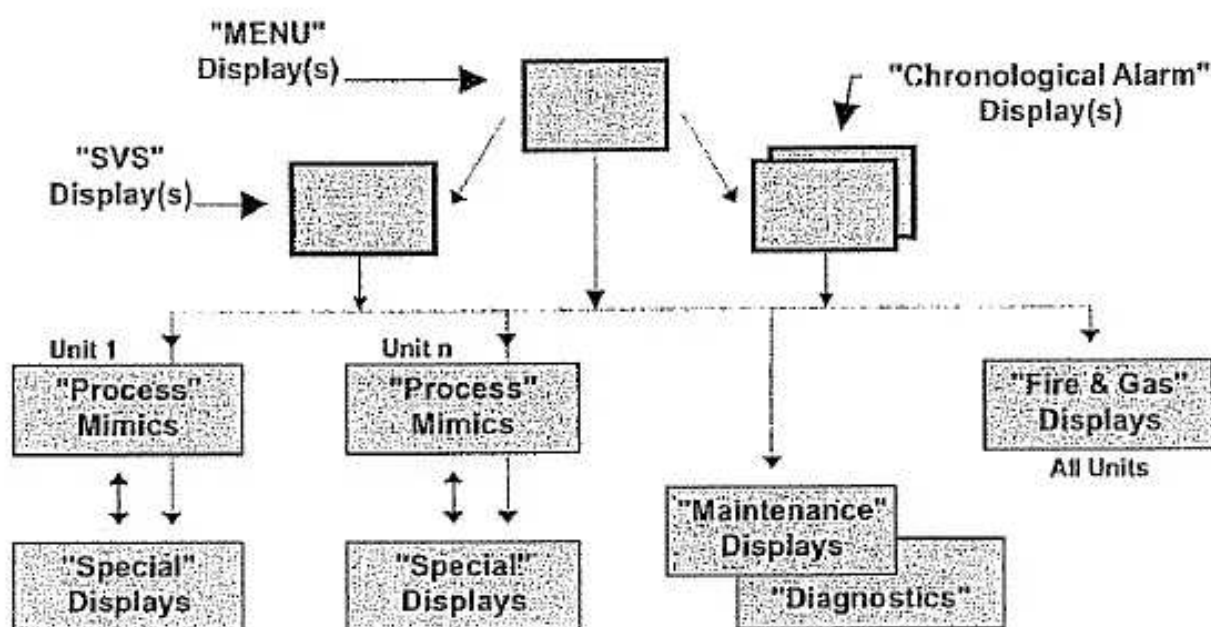
This example shows the hierarchy adopted by Lindsey Oil Refinery. The control desk for each operator will have 3 double stacked screens. The top left and top right screens display Spatial Vision Schematics (SVS), which are a form of simplified P&ID. The top middle screen displays chronological alarm pages. The top screen displays are normally fixed and cannot be changed. The bottom three screens are used for control. The system is display format driven as opposed to faceplate group displays.

Control Desk Layout



A Spatial Vision Schematic (SVS) is a Summary Display, giving a plant overview in the form of a Display format, rather than an Alphanumeric Display. It gives access to Display formats from which the plant can be controlled and surveyed in more detail. An SVS consists of Icons joined together by lines representing process streams. It is a simplified P&ID. There are two types of icon, Equipment (Pumps, Compressors, etc.) and Process (Heaters, Columns, etc.) Depending on the state of the process, the Icons are animated: they change colour and/or flash.

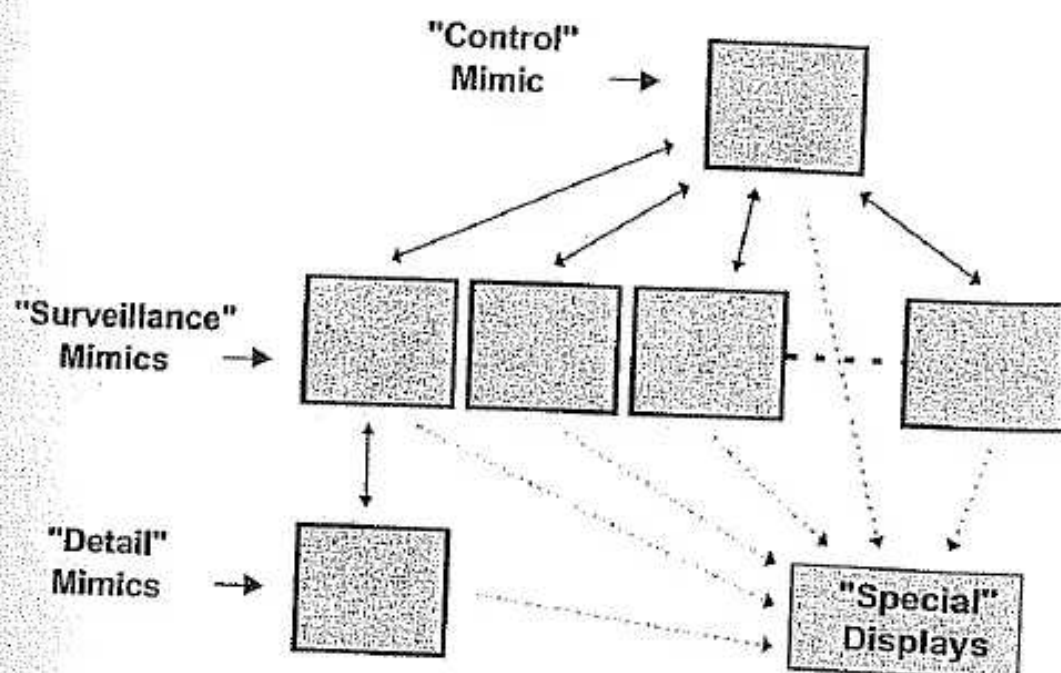
Display Hierarchy



The system incorporates different types of display formats :

Control Display format (for abnormal situations)	-Typical control and surveillance elements for Plant upset.
Surveillance Display format	Complete view of part of process covered.
Detail Display format	Information less relevant to direct operation of plant.

"Process" Mimics Hierarchy



Power plant generic example

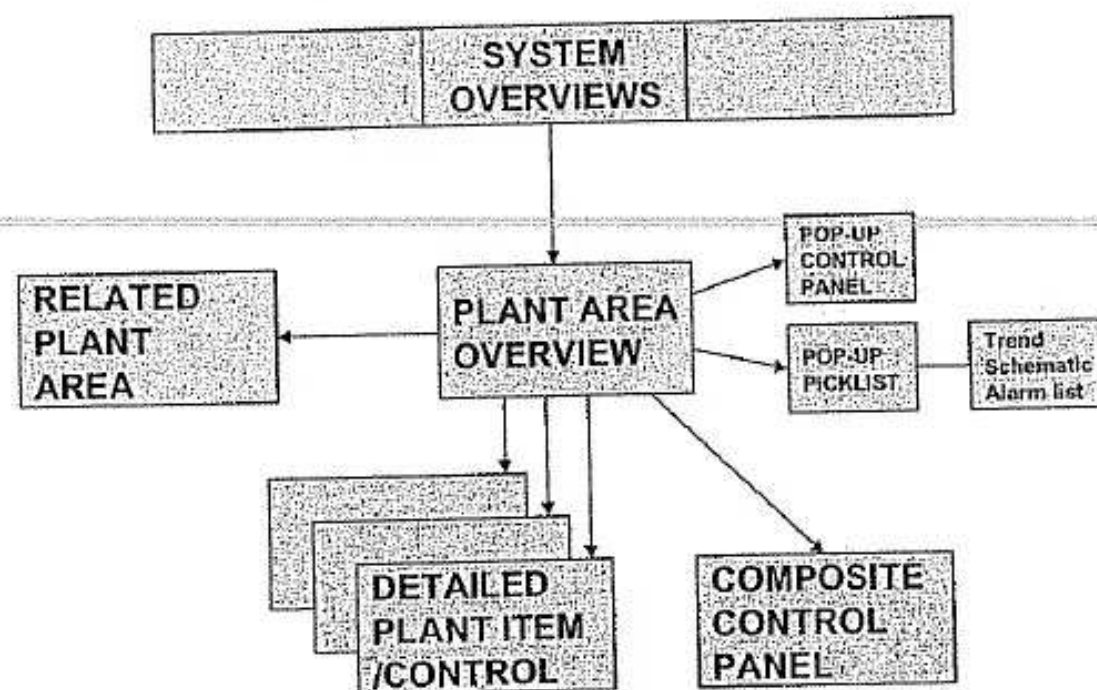
A number of conventional coal fired UK power plants have refurbished their HCI during the late 1990's. The same top end system (the Advanced Plant Management System) has been used in these refurbishments and a common approach has been taken with regard to the display hierarchy and navigation. The sites have generally installed fixed position display units to give overview displays of the three main plant areas - the boiler, the feed system and the turbine. A single display format for each area is left permanently on display, typically on large back projected screens.

The hierarchy below this level tends to be fairly flat. A number of more detailed plant area display formats were developed, each one focussing on a part of the overview plant area, with typically 'sideways' navigation between display formats within the same main overview plant area. For each of these display formats a number of lower level display formats were developed, with direct access from the 'parent' plant area display format. These would either give more detail of the plant or control of a particular plant item or be a composite plant item control display format.

In both the plant-area display formats and the more detailed display formats, both pop-up panels and pop-up menu lists could be accessed.

A number of the navigational techniques described earlier in the document were also utilised. The diagram below shows the hierarchical relationships between the display formats

Display Formats - Relationships



Explanation of display hierarchy levels:

Menu: the menu for each plant (or process) is accessible from the window toolbar. This menu gives single click access to all displays for the process and is arranged in a logical hierarchy as described below.

Note that an additional Menu page may be used for large processes.

- **Plant Overview:** provides an overview of the whole plant on a single page, only key vessels and indications such as main flow inputs/outputs are shown. Clicking on a vessel will call up the associated Detail graphic.
- **Unit Overview:** provides an overview of a single plant unit such as a column or reactor, a greater level on detail is shown compared to the Plant Overview. The operator can page through the Unit Overviews at this level by using the forward/backward icons or click on a vessel to call up the associated Detail graphic.
- **Detail:** provides a detailed view of the plant showing all analogue and discrete indications and controllers. There may be several Detail graphics for each Unit and again the operator can page through using the forward/backward icons. The operator is only permitted to make a set-point change at this level where he has the advantage of seeing all related measurements and can best assess the impact on the process of the change he is about to make.
- **ESD:** dedicated graphics that show cause and effect information for all trips associated with the DCS and independent trip system. The intent is to remove superfluous information so that the operator can quickly confirm what caused the trip and that the tripped items have gone to the correct failsafe states. Any items that do not reach the failsafe-state are highlighted accordingly.
- **Overlay:** A pop-up window that displays detailed information for the selected point and allows the operator to make set-point or other parameter changes
- **Trend:** A pop-up window that shows the trended history for the selected point.

Appendix 4.

Example refurbishment design strategy

This appendix outlines the processes that need to be followed in creating a new soft-desk control desk for an existing installation.

When refurbishing an existing plant and upgrading from a hard-desk to a soft-desk, some advantage can be taken of the fact that the operating mode of the plant is well known and the use of the indications well understood. This gives a basis for developing the new soft-desk display format hierarchy and navigation. It can be very beneficial to question the existing methods of performing tasks - i.e. why is the task performed this way, do we need all these indications, are they actually used etc. This can lead to a simplification of the process or at least a better understanding. An additional approach can also be to define new ways of approaching tasks, so that the soft-desk does not become just a replica of the hard-desk. The opportunities offered by the technology can thus be exploited.

An outline of the process that can be adopted in moving from a hard to soft-desk follows :

- Collate Existing Information
 - Obtain current panel information, controls, indicators
 - Obtain current VDU display formats - if any exist
 - Obtain current alarm information
 - List emergency scenarios and collect information on existing procedures

This information will enable a proper analysis of the requirements to be performed. The use and reason for indications can be questioned and a checklist of all required indications produced to ensure that the new desk covers all requirements. Knowledge of existing procedures will allow for task based display formats to be developed or new ways of displaying the required information to be formulated.

- Design HCI standard
 - Include guidelines on aspects as detailed in section 3 (Display format design guidance) of this guide.
- Design Overview Displays
 - Analyse information required
 - Limit to major indicators giving overall plant status (e.g. safety, efficiency, environmental)
 - Indicate major plant in/out of service
 - Consider major alarm information
 - Design for graphic, rather than numeric indication

Information collected in the previous step is crucial for this stage.

- Define overall Hierarchy of system
 - Define all major plant area formats required
 - Define all lower level more detail display formats
 - Define control action display formats required
 - Define navigation between display formats

The procedure uses a top down methodology, understanding the 'bigger picture' before moving down to see the detail that is required.

- Design Display formats
 - Analyse information required on each display format
 - Analyse auxiliary plant & services related to this plant item
 - Check alarm indication required
 - Identify button access to related display formats

At this point individual display format design is important with an emphasis on clarity and consistency. The navigational methods must be incorporated onto the screens.

- Design Control display formats
 - Identify all control actions needed for particular plant items
 - Identify requirements for 'group' control
 - Identify any control actions that are difficult to translate from hard-desk (e.g. two-handed operations, parallel operation)
- Design Trend displays
 - Collate information about existing chart recorders
 - Identify additional trend requirements
 - Define variables, timebase, scales
- Define alarms
 - Consult EEMUA alarm guidance document (ref. 3)

The design of alarms is a huge topic in its own right. The document referenced provides invaluable guidance on this issue. It must be recognised that alarm system design can be time and resource intensive, however if a HCI is to be the only operator interface then it is crucial for this to be performed properly.

- Build complete system (or prototype a small area)
 - To allow for review
- Review for acceptance
 - Check against designs
 - Check guidance conformity
 - Check pop-ups
 - Check control actions
 - Check display linkage

This formal off-line testing can result in significantly less problems at the commissioning stage.

- Commission
 - Check formats against actual plant-state
 - Check controls result in desired action
 - Check alarms are correctly connected to display

Appendix 5.**Engineer survey form**

This appendix shows a form that may be used to assess a Control & Instrumentation engineer's views on existing HCI on their plant. This can be a useful method of determining where problems exist in a HCI.

Survey of Process Systems Human Computer Interfaces (HCI) Questionnaire for Technical Staff

Insert here text explaining the questionnaire. This should typically include :

- Why the survey is being carried out*
- What is the purpose of the survey*
- Whether the survey is confidential*
- Who is carrying out the survey*
- Where to return the survey forms*

The following is example text:

This questionnaire forms part of a study aimed at understanding the issues associated with using 'soft-desks' - i.e. process control systems operator desks that are fully computerised and just use computer displays for displaying information to the operator. In particular it is concerned with the number of display units available to the operator and the ease with which the operator can navigate through the system in order to retrieve information. All information received through this survey will remain anonymous.

This questionnaire is aimed at the broader picture - trying to understand the type of plant that the operator has to deal with and some of the managerial and technical issues associated with it. A related operator's questionnaire should also be filled out for each plant - by as many operators as practicable.

If there is not enough room for full answers in any of the boxes, please continue on a separate sheet of paper. Also, any extra comments or information that you may wish to add will be gratefully received.

Many thanks for your time and understanding.

Company :	Site :
	Date :
1. What is your job and on what plant/unit ?	

2. What types of Process Control Systems are installed at this location?

3. What type of plant is this ? (e.g. for power generation - CHP, CCGT, coal etc)

4. What is the size and complexity of the plant ?

5. What level of manning & types of manning regime's are present at this plant ? - e.g. single/multiple manning, types of shifts etc

6. What level of automation is installed on the plant ? How does this affect the amount of operation required of the operator ?

7. What is the operating regime of the plant - e.g. continuously on, batch with start-ups periodically, continually stopping and starting etc.

8. Are control desks dedicated to single units ?

If not what sort of interaction is there ?

9. Are any control desks dedicated to providing common services ?

If so how are the screens delimited between units ?

10. What other systems are available to the operator in abnormal situations ?

11. How do the systems cope with screen failures ? i.e. what is the redundancy capability

12. What information do you think should be permanently on view to the operator and how many screens would this require ?

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13. Do you think having large overview screens would help you ? Comments:	Yes	No
14. Could the navigation around the system be improved ? If so, how ?	Yes	No
15. Would you prefer to use a different device to a 'mouse' ?	Yes	No

If you have any other comments on screen number, display formats or navigating through the display hierarchy, please write them on the blank sheet below:

Appendix 6.**Operator survey form**

This appendix shows a form that may be used to assess Operator's views on their existing HCI. This can be a useful method of determining where problems exist in a HCI.

**Survey of Process Systems Human Computer
Interfaces (HCI)
Questionnaire for Operators**

Insert here text explaining the questionnaire. This should typically include :

Why the survey is being carried out

What is the purpose of the survey

Whether the survey is confidential

Who is carrying out the survey

Where to return the survey forms

The following is example text:

This questionnaire forms part of a study aimed at understanding the issues associated with using 'soft-desks' - i.e. process control systems operator desks that are fully computerised and just use computer displays for displaying information to the operator. In particular it is concerned with the number of display units available to the operator and the ease with which the operator can navigate through the system in order to retrieve information. All information received through this survey will remain anonymous.

This questionnaire is aimed specifically at your impressions from operating the plant. It is concerned with the number of physical display screens that are available, navigation around the display format hierarchy and the usability of the system.

Any extra comments or information that you may wish to add will be gratefully received.

Many thanks for your time and understanding.

Company :	Site :
	Date :

1. What is your job and on what plant/unit ?		
2. How long have you worked on this Process Control System ?	___ Yrs mnths	
3. How many physical screens are available to you ?	___ Screens	
4. Is the physical arrangement of the screens satisfactory ? If not, how could it be improved ?	Yes	No
5. How many individual graphic layout display formats are available on the system ?	___ Display formats	
6. How many screens do you usually keep with the same fixed display format and how many do you vary ?	___ Fixed	___ Variable
7. How many screens do you typically, actively use :		
When the plant is in steady state ?	Screens	
When the plant is starting-up ?	Screens	
When the plant is in an abnormal situation ? (e.g. trip)	Screens	
8. Do you find it easy to navigate through the screen hierarchy ? Any Comments ?	Yes	No
9. How many operations (e.g. 'mouse clicks') does it typically take for you to get to the display format you wish to view ?	___ Operations	

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10 Can you display all the information that you need to do your job ? In steady state conditions In start-up conditions In abnormal situations (e.g. trip)	Yes Yes Yes	No No No
11 In abnormal situations, can you navigate through the format hierarchy quickly enough ?	Yes	No
12 Do you ever 'get lost' in the display format hierarchy ?	Yes	No
13. Is the amount of information displayed on each display format : Too much ____ About right ____ Too little ____		
14. Would having more physical screens help ?	Yes	No
15. Does your system have large overview displays ?	Yes	No
16. If 'NO to Question 15' then - Do you think having large overview screens would help you ?	Yes	No
17. Could the navigation around the system be improved ? If so, how ?	Yes	No
18. Would you prefer to use a different device to a 'mouse' ?	Yes	No

If you have any other comments on screen number, display formats or navigating through the display hierarchy, please write them on the blank sheet below:

Appendix 7.**References**

The appendix provides a bibliography of literature referred to in this report

- 1 ASM Consortium "Effective Operator Display Design Guidelines" Version 2.01 Honeywell Technology Centre, 3660 Technology Drive, Minneapolis, MN 55418 To be published.
- 2 BS "Functional Safety of Electrical/ Electronic/ Programmable Electronic Safety Related Systems" BS IEC 61508
- 3 EEMUA "Alarm Systems: A guide to design, management and procurement" EEMUA publication no 191, 1999.
- 4 Health & Safety Executive "The explosion and fires at Texaco Refinery, Milford Haven, 24th July 1994", HSE, 1997.
- 5 ISO "Ergonomic Design of Control Centres: Part 5 Displays and Controls" ISO 11064, under development.
- 6 Kirwan, B & Ainsworth, L. (eds) "A Guide to Task Analysis" 1994

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| 153 : 96 | EEMUA Supplement to ASME B31.3 —1996 Edition, Process Piping |
| 164 : 93 | Seal-less Centrifugal Pumps: Class 1 |
| 168 : 91 | A Guide to the Pressure Testing of In-Service Pressurised Equipment |
| 170 : 91 | Specification for Production Testing of Valves — Part 1 Ball Valves |
| 171 : 94 | Specification for Production Testing of Valves — Part 2 Plug Valves |
| 172 : 95 | Specification for Production Testing of Valves — Part 3 Gate Valves |
| 173 : 95 | Specification for Production Testing of Valves — Part 4 Butterfly and Globe Valves |
| 205 : 06* | Guide to the Production Testing of Valves for the Energy, Process, Oil and Gas Industries <i>(formerly Pubs 170, 171, 172 and 173 combined in a single publication)</i> |
| 179 : 96 | A Working Guide for Carbon Steel Equipment in Wet H ₂ S Service <i>(Developed largely from Oil Refinery experience)</i> |
| 182 : 04 | Specification for Integral Block and Bleed Valve Manifolds for Direct Connection to Pipework <i>(2nd edition, incorporating Corrigenda Jun 04)</i> |
| 184 : 96 | Guide to the Isolation of Pressure Relieving Devices |
| 185 : 96 | Guide for Hot Tapping on Piping and other Equipment <i>(Reprinted 2000 incorporating Amdts No 1 Dec 97)</i> |
| 188 : 99 | Guide for Establishing Operating Periods of Safety Valves |
| 192 : 98 | Guide for the Procurement of Valves for Low Temperature (Non-cryogenic) Service |
| 196 : 99 | Valve Purchasers' Guide to the European Pressure Equipment Directive <i>(Incorporating Amdts 1-3: Nov 00, Feb 02 & Aug 02)</i> |
| 199 : 00 | On-Line Leak Sealing of Piping - Guide to Safety Considerations |
| 200 : 00 | Guide to the Specification, Installation and Maintenance of Spring Supports for Piping |

- 203 : 04 Guide to the Application of ISO 3183 Parts 2 (1996) and 3 (1999) Petroleum and Natural Gas Industries — Steel Pipes for Pipelines — Technical Delivery Conditions
- 204 : 05 Piping and the European Pressure Equipment Directive: Guidance for Plant Owners/Operators (*Version 1: February 2005*)
- 208* Guide to the Life Cycle Management of Pressure Relief Systems

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- 144 : 87 90/10 Copper Nickel Alloy Piping for Offshore Applications — Specification: Tubes Seamless and Welded
- 145 : 87 90/10 Copper Nickel Alloy Piping for Offshore Applications — Specification: Flanges Composite and Solid
- 146 : 87 90/10 Copper Nickel Alloy Piping for Offshore Applications — Specification: Fittings
- 158 : 94 Construction Specification for Fixed Offshore Structures in the North Sea (*Reprinted 2005 incorporating Amdts 1-4: Apr 97 (1&2), Sept 00 & Mar 05*)
- 176 : 98 Specification for Structural Castings for Use Offshore
- 194 : 04 Guidelines for Materials Selection and Corrosion Control for Subsea Oil and Gas Production Equipment (*2nd edition*)
- 197 : 99 Specification for the Fabrication of Non-Primary Structural Steelwork for Offshore Installations

STORAGE TANKS AND VESSELS

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- 154 : 02 Guidance to Owners on Demolition of Vertical Cylindrical Steel Storage Tanks and Storage Spheres (*2nd edition*)
- 159 : 03 Users' Guide to the Inspection, Maintenance and Repair of Above-ground Vertical Cylindrical Steel Storage Tanks (*3rd edition incorporating Corrigenda & Amdts 1, Feb 04 to Vol 1, and Amdts 2, Jan 05 to Vol 2*)
- 180 : 96 Guide for Designers and Users on Frangible Roof Joints for Fixed Roof Storage Tanks
- 183 : 99 Guide for the Prevention of Bottom Leakage from Vertical Cylindrical Steel Storage Tanks (*Amdt Jan 04*)
- 190 : 00 Guide for the Design, Construction and Use of Mounded Horizontal Cylindrical Steel Vessels for Pressurised LPG at Ambient Temperatures (*Reprinted 2005 incorporating Errata, Oct 03, Amdts 1, Nov 04, and further amendments*)

207* Recommendations on the Design, Construction and Operation of Double Concrete Tanks for Refrigerated Storage

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140 : 85 Noise Procedure Specification

141 : 85 Guide to the Use of Noise Procedure Specification

161 : 85 Guide to the Selection and Assessment of Silencers and Acoustic Enclosures

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148 : 86 Reliability Specification - Model clauses for inclusion in purchasing specifications for equipment items and packages

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193 : 99 Recommendations for the Training, Development and Competency Assessment of Inspection Personnel

~~195 : 99 Compendium of EEMUA Information Sheets on Topics Related to Pressure Containing Equipment~~

206 : 06 Risk Based Inspection – A Guide to Effective Use of the RBI Process